

11

Infinite Series

Infinite series are useful in advanced courses in mathematics, physics, and engineering because they may be employed to represent functions in a new way. In this

chapter we shall discuss some of the fundamental results associated with this important mathematical concept.

11.1 Infinite Sequences

A function f from a set X to a set Y is a correspondence that associates with each element x of X a unique element $f(x)$ of Y (see Definition (1.21)). Until now the domain X has usually been an interval of real numbers. In this section we shall consider another type of function.

Definition (11.1)

An **infinite sequence** is a function whose domain is the set of positive integers.

For convenience we sometimes refer to infinite sequences merely as *sequences*. In this book the range of an infinite sequence will be a set of real numbers.

If f is an infinite sequence, then to each positive integer n there corresponds a real number $f(n)$. These numbers in the range of f may be listed by writing

$$f(1), f(2), f(3), \dots, f(n), \dots$$

where the dots at the end indicate that the sequence does not terminate. The number $f(1)$ is called the **first term** of the sequence, $f(2)$ the **second term** and, in general, $f(n)$ the **n th term** of the sequence. It is customary to use a subscript notation and write these numbers as follows:

$$(11.2) \quad a_1, a_2, a_3, \dots, a_n, \dots$$

In (11.2) it is understood that for each positive integer n , the symbol a_n denotes the real number $f(n)$. In this way we obtain an infinite collection of real numbers which is *ordered* in the sense that there is a first number, a second number, a forty-fifth number, and so on. Although sequences are functions, a collection such as (11.2) will also be referred to as an infinite sequence. If we wish to convert (11.2) to a function f , we let $f(n) = a_n$ for each positive integer n .

From the definition of equality of functions we see that a sequence

$$a_1, a_2, a_3, \dots, a_n, \dots$$

is **equal** to a sequence

$$b_1, b_2, b_3, \dots, b_n, \dots$$

if and only if $a_i = b_i$ for every positive integer i . Infinite sequences are often defined by stating a formula for the n th term, as in the next example.

Example 1 List the first four terms and the tenth term of the sequence whose n th term a_n is as follows.

$$\begin{array}{ll} \text{(a)} \quad a_n = \frac{n}{n+1} & \text{(b)} \quad a_n = 2 + (0.1)^n \\ \text{(c)} \quad a_n = (-1)^{n+1} \frac{n^2}{3n-1} & \text{(d)} \quad a_n = 4 \end{array}$$

Solution To find the first four terms we substitute, successively, $n = 1, 2, 3$, and 4 in the formula for a_n . The tenth term is found by substituting 10 for n . Doing this and simplifying gives us the following:

n th term	First four terms	Tenth term
(a) $\frac{n}{n+1}$	$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$	$\frac{10}{11}$
(b) $2 + (0.1)^n$	2.1, 2.01, 2.001, 2.0001	2.0000000001
(c) $(-1)^{n+1} \frac{n^2}{3n-1}$	$\frac{1}{2}, -\frac{4}{5}, \frac{9}{8}, -\frac{16}{11}$	$-\frac{100}{29}$
(d) 4	4, 4, 4, 4	4

■

We sometimes denote the sequence (11.2) by $\{a_n\}$. For example, the sequence $\{2^n\}$ has n th term $a_n = 2^n$. According to Definition (11.1), the sequence $\{2^n\}$ is the function f such that $f(n) = 2^n$ for every positive integer n .

Some infinite sequences $\{a_n\}$ have the property that as n increases, a_n gets very close to some real number L . Another way of stating this is to say that $|a_n - L|$ is almost zero when n is large. As an illustration, consider the sequence $\{a_n\}$ where

$$a_n = 2 + \left(-\frac{1}{2}\right)^n.$$

The first few terms are

$$2 - \frac{1}{2}, 2 + \frac{1}{4}, 2 - \frac{1}{8}, 2 + \frac{1}{16}, 2 - \frac{1}{32}, \dots$$

and it appears that the terms get closer to 2 as n increases. As a matter of fact, for every positive integer n ,

$$|a_n - 2| = \left| 2 + \left(-\frac{1}{2}\right)^n - 2 \right| = \left| \left(-\frac{1}{2}\right)^n \right| = \left(\frac{1}{2}\right)^n = \frac{1}{2^n}$$

and the number $1/2^n$, and hence $|a_n - 2|$, can be made arbitrarily close to zero by choosing n sufficiently large. According to the next definition, the given sequence has the limit 2, and we write

$$\lim_{n \rightarrow \infty} \left[2 + \left(-\frac{1}{2}\right)^n \right] = 2.$$

The situation here is almost identical to that in Chapter 4, where $\lim_{x \rightarrow \infty} f(x) = L$ was defined. The only difference is that if $f(n) = a_n$, then the domain of f is the set of positive integers and not an infinite interval of real numbers. As in Definition (4.19), but using a_n instead of $f(x)$, we state the following definition.

Definition (11.3)

A sequence $\{a_n\}$ has the limit L , written

$$\lim_{n \rightarrow \infty} a_n = L$$

if for every $\varepsilon > 0$, there exists a positive number N such that

$$|a_n - L| < \varepsilon \quad \text{whenever } n > N.$$

If $\lim_{n \rightarrow \infty} a_n$ does not exist in the sense of Definition (11.3), then the sequence $\{a_n\}$ has no limit.

A geometric interpretation similar to that shown in Figure 4.29 can be given for the limit of a sequence. The only difference is that the coordinate x of the point shown on the x -axis is always a positive integer. In Figure 11.1 we have illustrated the behavior of the points (k, a_k) for a special case in which $\lim_{n \rightarrow \infty} a_n = L$. Note that for any $\varepsilon > 0$, the points (n, a_n) lie between the lines $y = L \pm \varepsilon$, provided n is sufficiently large. Of course, the approach to L may vary from that illustrated in the figure.

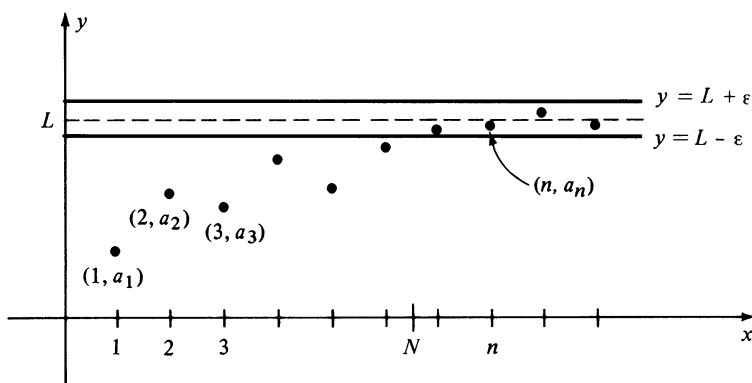


FIGURE 11.1

Another geometric interpretation of Definition (11.3) may be obtained by plotting the points corresponding to a_k on a coordinate line, as shown in Figure 11.2. In this case if we consider any open interval $(L - \varepsilon, L + \varepsilon)$, then a_n is in the interval whenever n is sufficiently large.

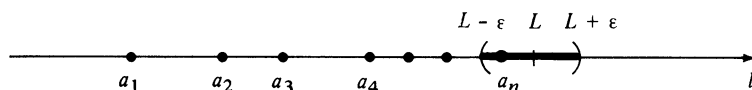


FIGURE 11.2

If a_n can be made as large as desired by choosing n sufficiently large, then the sequence $\{a_n\}$ has no limit, but we still write $\lim_{n \rightarrow \infty} a_n = \infty$. A more precise way of specifying this behavior is as follows.

Definition (11.4)

The statement $\lim_{n \rightarrow \infty} a_n = \infty$ means that for every positive real number P , there exists a number N such that $a_n > P$ whenever $n > N$.

The proof of the next theorem illustrates the use of the preceding definitions.

Theorem (11.5)

- (i) $\lim_{n \rightarrow \infty} r^n = 0$ if $|r| < 1$
- (ii) $\lim_{n \rightarrow \infty} |r^n| = \infty$ if $|r| > 1$

Proof If $r = 0$ it follows trivially that the limit is 0. Let us assume that $0 < |r| < 1$. To prove (i) by means of Definition (11.3) we must show that for every $\varepsilon > 0$ there exists a positive number N such that

$$\text{if } n > N, \text{ then } |r^n - 0| < \varepsilon.$$

The inequality $|r^n - 0| < \varepsilon$ is equivalent to each inequality in the following list:

$$|r|^n < \varepsilon, \quad n \ln |r| < \ln \varepsilon, \quad n > \frac{\ln \varepsilon}{\ln |r|}$$

where the final inequality sign is reversed because $\ln |r|$ is negative if $0 < |r| < 1$. The last inequality in the list gives us a clue to the choice of N . Specifically, if $\varepsilon < 1$, then $\ln \varepsilon < 0$ and we let $N = \ln \varepsilon / \ln |r| > 0$. In this event if $n > N$, then the last inequality in the list is true, and hence so is the first, which is what we wished to prove. If $\varepsilon \geq 1$, then $\ln \varepsilon \geq 0$ and hence $\ln \varepsilon / \ln |r| \leq 0$. In this case if N is any positive number, then whenever $n > N$ the last inequality in the list is again true.

To prove (ii) let $|r| > 1$ and consider any positive real number P . The following inequalities are equivalent:

$$|r|^n > P, \quad n \ln |r| > \ln P, \quad n > \frac{\ln P}{\ln |r|}.$$

If we choose $N = \ln P / \ln |r|$, then whenever $n > N$ we have $|r|^n > P$. Hence by Definition (11.4), $\lim_{n \rightarrow \infty} |r|^n = \infty$. $\square$

Example 2 List the four terms and, if possible, find the limits of each of the following sequences.

(a) $\{(-\frac{2}{3})^n\}$ (b) $\{(1.01)^n\}$

Solution

(a) The first four terms of $\{(-\frac{2}{3})^n\}$ are

$$-\frac{2}{3}, \quad \frac{4}{9}, \quad -\frac{8}{27}, \quad \frac{16}{81}.$$

According to Theorem (11.5),

$$\lim_{n \rightarrow \infty} \left(-\frac{2}{3}\right)^n = 0.$$

(b) The first four terms of $\{(1.01)^n\}$ are

$$1.01, \quad 1.0201, \quad 1.030301, \quad 1.04060401.$$

According to Theorem (11.5),

$$\lim_{n \rightarrow \infty} (1.01)^n = \infty. \quad \blacksquare$$

Theorems that are analogous to some of the limit theorems in Chapter 2 may be established for infinite sequences. In particular, given two infinite sequences $\{a_n\}$ and $\{b_n\}$, corresponding terms may be added to form the infinite sequence $\{a_n + b_n\}$, which is called the **sum** of the given sequences. Similarly, we could form the **difference**, **product**, or **quotient** by subtracting,

multiplying, or dividing corresponding terms (provided no zero denominators occur in the division process).

If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$ where L and M are real numbers, then it can be shown that

$$\begin{aligned} \lim_{n \rightarrow \infty} (a_n + b_n) &= L + M, & \lim_{n \rightarrow \infty} (a_n - b_n) &= L - M, \\ \lim_{n \rightarrow \infty} a_n b_n &= LM, & \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \frac{L}{M}. \end{aligned}$$

In the last formula we must have $M \neq 0$ and $b_n \neq 0$ for all n .

If $a_n = c$ for all n , so that the infinite sequence is $c, c, \dots, c, \dots$, then

$$\lim_{n \rightarrow \infty} c = c.$$

Similarly, if c is a real number and k is a positive rational number, then, as in the proof of Theorem (4.21),

$$\lim_{n \rightarrow \infty} \frac{c}{n^k} = 0.$$

Example 3 Find the limit of the sequence $\left\{ \frac{2n}{5n-3} \right\}$.

Solution We wish to find $\lim_{n \rightarrow \infty} a_n$, where $a_n = 2n/(5n-3)$. Dividing numerator and denominator of a_n by n and applying limit theorems, we obtain

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2n}{5n-3} &= \lim_{n \rightarrow \infty} \frac{2}{5 - (3/n)} = \frac{\lim_{n \rightarrow \infty} 2}{\lim_{n \rightarrow \infty} [5 - (3/n)]} \\ &= \frac{2}{\lim_{n \rightarrow \infty} 5 - \lim_{n \rightarrow \infty} (3/n)} = \frac{2}{5 - 0} = \frac{2}{5}. \end{aligned}$$

Hence the sequence has the limit $\frac{2}{5}$. ■

Let $\{a_n\}$ be a sequence and consider the function f , where $f(n) = a_n$. It is often true that $f(x)$ is defined for every *real* number $x \geq 1$. In this case it follows from Definitions (11.3) and (4.19) that

$$\text{if } \lim_{x \rightarrow \infty} f(x) = L, \text{ then } \lim_{n \rightarrow \infty} f(n) = L.$$

This fact enables us to apply theorems about limits of functions (as $x \rightarrow \infty$) to limits of sequences. Of special importance is L'Hôpital's Rule, as illustrated in the next example.

Example 4 Find the limit of the sequence $\{5n/e^{2n}\}$.

Solution Let $f(x) = 5x/e^{2x}$ where x is real. Since f takes on the indeterminate form ∞/∞ as $x \rightarrow \infty$, we may use L'Hôpital's Rule, obtaining

$$\lim_{x \rightarrow \infty} \frac{5x}{e^{2x}} = \lim_{x \rightarrow \infty} \frac{5}{2e^{2x}} = 0.$$

Hence, by the discussion preceding this example,

$$\lim_{n \rightarrow \infty} \frac{5n}{e^{2n}} = 0. \quad \blacksquare$$

The next theorem is analogous to Theorem (2.14). It states that if the terms of an infinite sequence are always sandwiched between corresponding terms of two sequences which have the same limit L , then the given sequence also has the limit L .

**The Sandwich Theorem
for Infinite Sequences**
(11.6)

If $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ are infinite sequences such that $a_n \leq b_n \leq c_n$ for all n , and if

$$\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n$$

then $\lim_{n \rightarrow \infty} b_n = L$.

Proof For every $\varepsilon > 0$ there corresponds a number M such that if $n > M$, then both $|a_n - L| < \varepsilon$ and $|c_n - L| < \varepsilon$; that is,

$$L - \varepsilon < a_n < L + \varepsilon \quad \text{and} \quad L - \varepsilon < c_n < L + \varepsilon.$$

Consequently, if $n > M$, then

$$L - \varepsilon < a_n \quad \text{and} \quad c_n < L + \varepsilon.$$

Since $a_n \leq b_n \leq c_n$, it follows that if $n > M$, then $L - \varepsilon < b_n < L + \varepsilon$ or, equivalently, $|b_n - L| < \varepsilon$. This completes the proof. $\square$

Example 5 Find the limit of the sequence $\left\{\frac{\cos^2 n}{3^n}\right\}$.

Solution Since $0 < \cos^2 n < 1$ for every positive integer n , we may write

$$0 < \frac{\cos^2 n}{3^n} < \frac{1}{3^n}.$$

Applying Theorem (11.5) with $r = \frac{1}{3}$,

$$\lim_{n \rightarrow \infty} \frac{1}{3^n} = \lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0.$$

Moreover $\lim_{n \rightarrow \infty} 0 = 0$. It follows from the Sandwich Theorem (11.6) with $a_n = 0$, $b_n = (\cos^2 n)/3^n$, and $c_n = (\frac{1}{3})^n$, that

$$\lim_{n \rightarrow \infty} \frac{\cos^2 n}{3^n} = 0.$$

Hence the limit of the sequence is 0. ■

The proof of the next theorem is left as an exercise.

Theorem (11.7)

Let $\{a_n\}$ be a sequence. If $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$.

Example 6 Suppose the n th term of a sequence is $a_n = (-1)^{n+1}(1/n)$. Prove that $\lim_{n \rightarrow \infty} a_n = 0$.

Solution The terms of the sequence are alternately positive and negative. For example, the first five terms are

$$1, \quad -\frac{1}{2}, \quad \frac{1}{3}, \quad -\frac{1}{4}, \quad \frac{1}{5}.$$

Since

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0,$$

it follows from Theorem (11.7) that $\lim_{n \rightarrow \infty} a_n = 0$. ■

A sequence is said to be **monotonic** if successive terms are nondecreasing in the sense that

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots$$

or nonincreasing in the sense that

$$a_1 \geq a_2 \geq \cdots \geq a_n \geq \cdots$$

A sequence is **bounded** if there is a positive real number M such that $|a_k| \leq M$ for all k . The next theorem is fundamental for later developments.

Theorem (11.8)

A bounded, monotonic, infinite sequence has a limit.

To prove Theorem (11.8), it is necessary to use an important property of real numbers. Let us first state several definitions. If S is a nonempty set of real numbers, then a real number u is called an **upper bound** of S if $x \leq u$ for

every x in S . A number v is a **least upper bound** of S if v is an upper bound and no number less than v is an upper bound of S . The least upper bound is, therefore, the smallest real number that is greater than or equal to every number in S . To illustrate, if S is the open interval (a, b) , then any number greater than b is an upper bound of S ; however, the least upper bound of S is unique, and equals b (see Exercise 50).

The following statement is an axiom for the real number system.

The Completeness Property (11.9)

If a nonempty set S of real numbers has an upper bound, then S has a least upper bound.

Proof of Theorem (11.8) Let $\{a_n\}$ be a bounded monotonic sequence with nondecreasing terms. Thus

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots$$

and there is a number M such that $a_k \leq M$ for every positive integer k . Since the set

$$S = \{a_k : k = 1, 2, \dots\}$$

has an upper bound, it follows from the Completeness Property that S has a least upper bound L , where $L \leq M$. If $\varepsilon > 0$, then $L - \varepsilon$ is not an upper bound of S and hence at least one term of $\{a_n\}$ is greater than $L - \varepsilon$; that is,

$$L - \varepsilon < a_N \quad \text{for some positive integer } N.$$

Since the terms of $\{a_n\}$ are nondecreasing, we have

$$a_N \leq a_{N+1} \leq a_{N+2} \leq \cdots$$

and, therefore,

$$L - \varepsilon < a_n \quad \text{for every } n > N.$$

It follows that if $n > N$, then

$$0 \leq L - a_n < \varepsilon, \quad \text{or} \quad |L - a_n| < \varepsilon.$$

By Definition (11.3) this implies that

$$\lim_{n \rightarrow \infty} a_n = L \leq M;$$

that is, $\{a_n\}$ has a limit. The proof for a sequence $\{a_n\}$ of nonincreasing terms may be obtained by a similar proof or by considering the sequence $\{-a_n\}$. □

We shall conclude this section with an application of sequences to the investigation of the time course of an $S \rightarrow I \rightarrow S$ epidemic.* Let us suppose that physicians issue daily reports indicating the number of persons who have become infected with the disease and those who have been cured. We shall label the reporting days as $1, 2, \dots, n, \dots$ and let I_n denote the number of individuals in a population of size N who have the disease on the n th day. Let F_n and C_n denote the number of newly infected and the number cured, respectively, on day n . Thus, for $n \geq 1$,

$$I_{n+1} = I_n + F_{n+1} - C_{n+1}.$$

Suppose health officials decide that the number of new cases on a given day is directly proportional to the product of the number ill and the number not infected on the previous day. (This is known as the *Law of Mass Action*, and is typical of a population consisting of students on a college campus.) In addition, assume that the number cured each day is directly proportional to the number ill the previous day. Hence,

$$F_{n+1} = aI_n(N - I_n) \quad \text{and} \quad C_{n+1} = bI_n$$

where a and b are constants. Substituting in the preceding formula for I_{n+1} , we obtain

$$I_{n+1} = I_n + aI_n(N - I_n) - bI_n.$$

In the early stages of an epidemic I_n will be very small compared to N , and from the point of view of public health, it is better to *overestimate* the number ill than it is to underestimate and be unprepared for the spread of the disease. With this in mind, the early dynamics of the epidemic may be realistically investigated by examining the equation

$$I_{n+1} = I_n + aNI_n - bI_n = (1 + aN - b)I_n$$

where I_1 is the initial number of infected individuals. If we let $r = 1 + aN - b$, then $I_{n+1} = rI_n$ and, therefore,

$$I_2 = rI_1, \quad I_3 = rI_2 = r^2I_1, \quad I_4 = rI_3 = r^3I_1, \quad \dots, \quad I_n = r^{n-1}I_1, \quad \dots$$

This gives us the following sequence of infected individuals:

$$I_1, rI_1, r^2I_1, r^3I_1, \dots, r^{n-1}I_1, \dots$$

The number $r = 1 + aN - b$ (which can be approximated from early data) is of critical import. If $r > 1$, then by Theorem (11.5), $\lim_{n \rightarrow \infty} I_n = \infty$ and an epidemic will be in progress. In this case, when n is large, I_n is no longer small compared to N , and the formula for I_{n+1} becomes invalid. If $r < 1$, then $\lim_{n \rightarrow \infty} I_n = 0$ and health officials need not be concerned. The case $r = 1$ results in the constant sequence $I_1, I_1, \dots, I_1, \dots$.

* The symbol $S \rightarrow I \rightarrow S$ is an abbreviation for *Susceptible* $\rightarrow$ *Infected* $\rightarrow$ *Susceptible*, and is used to signify that an infected person who becomes cured is not immune to the disease, but may contract it again. Examples of such diseases are gonorrhea or strep throat.

11.1 Exercises

Given the n th term a_n of an infinite sequence as in Exercises 1–16, find the first four terms and $\lim_{n \rightarrow \infty} a_n$, if it exists.

- | | |
|---|--|
| 1 $a_n = \frac{n}{3n+2}$ | 2 $a_n = \frac{6n-5}{5n+1}$ |
| 3 $a_n = \frac{7-4n^2}{3+2n^2}$ | 4 $a_n = \frac{4}{8-7n}$ |
| 5 $a_n = -5$ | 6 $a_n = \sqrt{2}$ |
| 7 $a_n = \frac{(2n-1)(3n+1)}{n^3+1}$ | 8 $a_n = 8n+1$ |
| 9 $a_n = \frac{4}{\sqrt{n^2+9}}$ | 10 $a_n = \frac{100n}{n^{3/2}+4}$ |
| 11 $a_n = (-1)^{n+1} \frac{3n}{n^2+4n+5}$ | 12 $a_n = (-1)^{n+1} \frac{\sqrt{n}}{n+1}$ |
| 13 $a_n = 1 + (0.1)^n$ | 14 $a_n = 1 - (1/2)^n$ |
| 15 $a_n = 1 + (-1)^{n+1}$ | 16 $a_n = (n+1)/\sqrt{n}$ |

In Exercises 17–42, find the limit of the sequence, if it exists.

- | | |
|--|--|
| 17 $\{6(-5/6)^n\}$ | 18 $\{8 - (7/8)^n\}$ |
| 19 $\{\arctan n\}$ | 20 $\{(\tan^{-1} n)/n\}$ |
| 21 $\{1000 - n\}$ | 22 $\{(1.0001)^n/1000\}$ |
| 23 $\left\{(-1)^n \frac{\ln n}{n}\right\}$ | 24 $\left\{\frac{n^2}{\ln(n+1)}\right\}$ |
| 25 $\left\{\frac{4n^4+1}{2n^2-1}\right\}$ | 26 $\left\{\frac{\cos n}{n}\right\}$ |
| 27 $\{e^n/n^4\}$ | 28 $\{e^{-n} \ln n\}$ |

- | | |
|---|--|
| 29 $\left\{\left(1 + \frac{1}{n}\right)^n\right\}$ | 30 $\{(-1)^n n^3 3^{-n}\}$ |
| 31 $\{2^{-n} \sin n\}$ | 32 $\left\{\frac{4n^3+5n+1}{2n^3-n^2+5}\right\}$ |
| 33 $\left\{\frac{n^2}{2n-1} - \frac{n^2}{2n+1}\right\}$ | 34 $\left\{n \sin \frac{1}{n}\right\}$ |
| 35 $\{\cos n\pi\}$ | 36 $\{4 + \sin(n\pi/2)\}$ |
| 37 $\{n^{1/n}\}$ | 38 $\{n^2/2^n\}$ |
| 39 $\left\{\frac{n^{-10}}{\sec n}\right\}$ | 40 $\left\{(-1)^n \frac{n^2}{1+n^2}\right\}$ |
| 41 $\{2^n/(2n)!\}$ | 42 $\{n[\ln(n+1) - \ln n]\}$ |
- 43 A test question lists the first four terms of a sequence as 2, 4, 6, and 8 and asks the person being tested to list the fifth term. Show that the fifth term can be any real number a by finding the n th term of a sequence that has for its first five terms 2, 4, 6, 8, and a .
- 44 If k is an integer greater than 1, prove that $\lim_{n \rightarrow \infty} 1/k^n = 0$. (*Hint*: Use Theorem (11.6).)
- 45 Prove that if $\{a_n\}$ has the limit L , then L is unique.
- 46 Prove, directly from Definition (11.3), that if $\lim_{n \rightarrow \infty} a_n = L$, then $\lim_{n \rightarrow \infty} |a_n| = |L|$.
- 47 Define, in a manner analogous to Definition (11.4), what is meant by $\lim_{n \rightarrow \infty} a_n = -\infty$.
- 48 Prove Theorem (11.7).
- 49 If $\{a_n\}$ has a limit and c is a real number, prove that $\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$.
- 50 Prove that b is the least upper bound of the open interval (a, b) .

11.2 Convergent or Divergent Infinite Series

Every rational number can be represented by a nonterminating, repeating decimal. For example, applying long division to $\frac{2}{3}$ we obtain the representation

$$\frac{2}{3} = 0.66666 \dots$$

This decimal may also be regarded as the limit of the sequence whose first few terms are

$$0.6, 0.66, 0.666, 0.6666, 0.66666, \dots$$

As we shall see, it is also possible to express $\frac{2}{3}$ as the *infinite sum* (or *series*)

$$0.6 + 0.06 + 0.006 + 0.0006 + 0.00006 + \dots$$

Since only finite sums may be added algebraically, it is necessary to introduce concepts that give meaning to an infinite sum of this type. In this section we shall state the required definitions; however, our development will take us far beyond this elementary application. Indeed, the major objective of this chapter is to represent *functions* by means of infinite sums.

Let us begin by introducing some terminology that will be used throughout the remainder of the chapter.

Definition (11.10)

Let $\{a_n\}$ be an infinite sequence. An expression of the form

$$a_1 + a_2 + \dots + a_n + \dots$$

is called an **infinite series**, or simply a **series**.

In summation notation, the series in Definition (11.10) will be denoted by either

$$\sum_{n=1}^{\infty} a_n \quad \text{or} \quad \sum a_n$$

where it is understood that the summation variable in the last sum is n . Each number a_i is called a **term** of the series, and a_n is called the **n th term**.

Definition (11.11)

(i) The **n th partial sum** S_n of the infinite series $\sum a_n$ is

$$S_n = a_1 + a_2 + \dots + a_n.$$

(ii) The **sequence of partial sums** associated with the infinite series $\sum a_n$ is

$$S_1, S_2, S_3, \dots, S_n, \dots$$

Thus, from (i) of Definition (11.11),

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$S_4 = a_1 + a_2 + a_3 + a_4.$$

To illustrate, consider the series

$$0.6 + 0.06 + 0.006 + \cdots + \frac{6}{10^n} + \cdots$$

The first few terms of the sequence of partial sums are

$$0.6, \quad 0.66, \quad 0.666, \quad 0.6666, \quad 0.66666, \quad \dots$$

which is the sequence considered at the beginning of this section. In Example 2 we will show that this sequence has the limit $\frac{2}{3}$. Since the partial sums S_n of the series approach $\frac{2}{3}$ as n increases, it is natural to refer to $\frac{2}{3}$ as the *sum* of the infinite series and write

$$\frac{2}{3} = 0.6 + 0.06 + 0.006 + \cdots + \frac{6}{10^n} + \cdots$$

The following definition extends these remarks to any infinite series.

Definition (11.12)

An infinite series

$$a_1 + a_2 + \cdots + a_n + \cdots$$

with sequence of partial sums $S_1, S_2, \dots, S_n, \dots$ **is convergent** (or **converges**) if $\lim_{n \rightarrow \infty} S_n = S$ for some real number S . The series **is divergent** (or **diverges**) if this limit does not exist.

If $a_1 + a_2 + \cdots + a_n + \cdots$ is a convergent infinite series and $\lim_{n \rightarrow \infty} S_n = S$, then S is called the **sum of the series**, and we write

$$S = a_1 + a_2 + \cdots + a_n + \cdots$$

If a series diverges, it has no sum.

Intuitively, if a series converges and has the sum S , then as we add more and more terms, the resulting partial sums get closer and closer to S .

Example 1 Prove that the infinite series

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)} + \cdots$$

converges and find its sum.

Solution The partial fraction decomposition of a_n is

$$a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Consequently, the n th partial sum of the series may be written

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \cdots + a_n \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} = \frac{n}{n+1}. \end{aligned}$$

Since
$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1,$$

the series converges and has the sum 1. ■

The series $\sum 1/[n(n+1)]$ of Example 1 is called a **telescoping series**, since writing S_n as shown in the solution of Example 1 causes the terms to collapse to $1 - 1/(n+1)$.

Certain types of infinite series arise frequently in applications. One example is a **geometric series**

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$$

where a and r are real numbers and $a \neq 0$. The next result is extremely important for work later in this chapter.

Theorem (11.13)

Let $a \neq 0$. The geometric series

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$$

- (i) converges and has the sum $\frac{a}{1-r}$ if $|r| < 1$.
- (ii) diverges if $|r| \geq 1$.

Proof If $r = 1$, then $S_n = a + a + \cdots + a = na$ and the series diverges, since $\lim_{n \rightarrow \infty} S_n$ does not exist.

If $r \neq 1$, consider

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}$$

and

$$rS_n = ar + ar^2 + ar^3 + \cdots + ar^n.$$

Subtracting corresponding sides of these equations we obtain

$$(1-r)S_n = a - ar^n.$$

Dividing both sides by $1-r$ gives us

$$S_n = \frac{a}{1-r} - \frac{ar^n}{1-r}.$$

Consequently,

$$\begin{aligned}\lim_{n \rightarrow \infty} S_n &= \lim_{n \rightarrow \infty} \left(\frac{a}{1-r} - \frac{ar^n}{1-r} \right) \\ &= \lim_{n \rightarrow \infty} \frac{a}{1-r} - \lim_{n \rightarrow \infty} \frac{ar^n}{1-r} \\ &= \frac{a}{1-r} - \frac{a}{1-r} \lim_{n \rightarrow \infty} r^n.\end{aligned}$$

If $|r| < 1$, then $\lim_{n \rightarrow \infty} r^n = 0$ (see Theorem (11.5)) and hence

$$\lim_{n \rightarrow \infty} S_n = \frac{a}{1-r}.$$

If $|r| > 1$, then $\lim_{n \rightarrow \infty} r^n$ does not exist (see Theorem (11.5)) and hence $\lim_{n \rightarrow \infty} S_n$ does not exist. In this case the series diverges. $\square$

Example 2 Prove that the infinite series

$$0.6 + 0.06 + 0.006 + \cdots + \frac{6}{10^n} + \cdots$$

converges and find its sum.

Solution The series is geometric with $a = 0.6$ and $r = 0.1$. By (i) of Theorem (11.13) it converges and has the sum

$$S = \frac{0.6}{1-0.1} = \frac{0.6}{0.9} = \frac{2}{3}. \quad \blacksquare$$

Example 3 Prove that the following series converges and find its sum:

$$2 + \frac{2}{3} + \frac{2}{3^2} + \cdots + \frac{2}{3^{n-1}} + \cdots$$

Solution The series converges since it is geometric with $r = \frac{1}{3} < 1$. By (i) of Theorem (11.13), the sum is

$$S = \frac{2}{1-\frac{1}{3}} = \frac{2}{\frac{2}{3}} = 3. \quad \blacksquare$$

Theorem (11.14)

If an infinite series $\sum a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof The n th term a_n of the infinite series can be expressed as

$$a_n = S_n - S_{n-1}.$$

If $\lim_{n \rightarrow \infty} S_n = S$, then also $\lim_{n \rightarrow \infty} S_{n-1} = S$ and

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (S_n - S_{n-1}) = \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} = 0. \quad \square$$

The preceding theorem states that *if* a series converges, *then* its n th term a_n has the limit 0 as $n \rightarrow \infty$. The converse is false, that is, *if* $\lim_{n \rightarrow \infty} a_n = 0$ *it does not necessarily follow* that the series $\sum a_n$ is convergent. (See Example 5 of this section.)

The next result is an immediate corollary of Theorem (11.14).

Test for Divergence (11.15)

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the infinite series $\sum a_n$ is divergent.

Example 4 Determine whether the following series converges or diverges:

$$\frac{1}{3} + \frac{2}{5} + \frac{3}{7} + \cdots + \frac{n}{2n+1} + \cdots$$

Solution Since

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2} \neq 0,$$

the series diverges by (11.15). ■

Theorem (11.16)

If an infinite series $\sum a_n$ is convergent, then for every $\varepsilon > 0$ there exists an integer N such that $|S_k - S_l| < \varepsilon$ whenever $k > N$ and $l > N$.

Proof If the sum of the series is S , then

$$|S_k - S_l| = |(S_k - S) + (S - S_l)| \leq |S_k - S| + |S - S_l|.$$

Since $\lim_{k \rightarrow \infty} S_k = S = \lim_{l \rightarrow \infty} S_l$, for every $\varepsilon > 0$ there is an integer N such that

$$|S_k - S| < \frac{\varepsilon}{2} \quad \text{and} \quad |S - S_l| < \frac{\varepsilon}{2}$$

whenever $k > N$ and $l > N$. Thus

$$|S_k - S_l| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \square$$

Example 5 Prove that the series $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$ is divergent.

Solution If $n > 1$, then

$$\begin{aligned} S_{2n} - S_n &= \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \\ &> \frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n} = \frac{1}{2}. \end{aligned}$$

If the series is convergent, then by Theorem (11.16) with $\varepsilon = \frac{1}{2}$, $k = 2n$, and $l = n$, $|S_{2n} - S_n| < \frac{1}{2}$ provided n is sufficiently large. Since the last inequality is *never* true, it follows that the series diverges. ■

The series in Example 5 will be important in later developments. Because of this we give it the following special name.

Definition (11.17)

The **harmonic series** is the divergent infinite series

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

The harmonic series is an illustration of a divergent series $\sum a_n$ for which $\lim_{n \rightarrow \infty} a_n = 0$. This shows that the converse of Theorem (11.14) is false. Consequently, to establish convergence of an infinite series *it is not enough* to prove that $\lim_{n \rightarrow \infty} a_n = 0$, since that may be true for divergent as well as convergent series.

The next theorem states that if corresponding terms of two infinite series are identical after a certain term, then both series converge or both series diverge.

Theorem (11.18)

If $\sum a_n$ and $\sum b_n$ are infinite series such that $a_i = b_i$ for all $i > k$, where k is a positive integer, then both series converge or both series diverge.

Proof By hypothesis we may write

$$\sum a_n = a_1 + a_2 + \cdots + a_k + a_{k+1} + \cdots + a_n + \cdots$$

and

$$\sum b_n = b_1 + b_2 + \cdots + b_k + a_{k+1} + \cdots + a_n + \cdots$$

Let S_n and T_n denote the n th partial sums of $\sum a_n$ and $\sum b_n$, respectively. It follows that if $n \geq k$, then

$$S_n - S_k = T_n - T_k$$

or

$$S_n = T_n + (S_k - T_k).$$

Consequently,

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} T_n + (S_k - T_k)$$

and hence either both of the limits exist or both do not exist. This gives us the desired conclusion. Evidently, if both series converge, then their sums differ by $S_k - T_k$. $\square$

Theorem (11.18) implies that changing a finite number of terms of an infinite series has no effect on its convergence or divergence (although it does change the sum of a convergent series). In particular, if we replace the first k terms of $\sum a_n$ by 0, convergence is unaffected. It follows that the series

$$a_{k+1} + a_{k+2} + \cdots + a_n + \cdots$$

converges or diverges according as $\sum a_n$ converges or diverges. The series $a_{k+1} + a_{k+2} + \cdots$ is said to have been obtained from $\sum a_n$ by **deleting the first k terms**.

Example 6 Show that the following series converges:

$$\frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \cdots + \frac{1}{(n+2)(n+3)} + \cdots$$

Solution The series converges since it can be obtained by deleting the first two terms of the convergent telescoping series of Example 1. $\blacksquare$

The proof of the next theorem follows directly from Definition (11.12) and is left as an exercise.

Theorem (11.19)

If $\sum a_n$ and $\sum b_n$ are convergent series with sums A and B , respectively, then

- (i) $\sum (a_n + b_n)$ converges and has sum $A + B$.
- (ii) if c is a real number, $\sum ca_n$ converges and has sum cA .
- (iii) $\sum (a_n - b_n)$ converges and has sum $A - B$.

It is also easy to show that if $\sum a_n$ diverges, then so does $\sum ca_n$ for every $c \neq 0$.

Example 7 Prove that the following series converges and find its sum:

$$\sum_{n=1}^{\infty} \left[\frac{7}{n(n+1)} + \frac{2}{3^{n-1}} \right].$$

Solution The telescoping series $\sum 1/[n(n+1)]$ was considered in Example 1, where it was shown that it converges and has the sum 1. Using (ii) of Theorem (11.19) with $c = 7$ and $a_n = 1/[n(n+1)]$, we see that $\sum 7/[n(n+1)]$ converges and has the sum $7(1) = 7$.

The geometric series $\sum 2/3^{n-1}$ converges and has the sum 3 (see Example 3). Hence by (i) of Theorem (11.19), the given series converges and has the sum $7 + 3$, or 10. ■

Theorem (11.20)

If $\sum a_n$ is a convergent series and $\sum b_n$ is divergent, then $\sum (a_n + b_n)$ is divergent.

Proof We shall give an indirect proof. Thus suppose $\sum (a_n + b_n)$ is convergent. Applying (iii) of Theorem (11.19), $\sum [(a_n + b_n) - a_n] = \sum b_n$ is convergent, a contradiction. Hence our supposition is false, that is, $\sum (a_n + b_n)$ is divergent. □

Example 8 Determine the convergence or divergence of

$$\sum_{n=1}^{\infty} \left(\frac{1}{5^n} + \frac{1}{n} \right).$$

Solution Since $\sum (1/5^n)$ is a convergent geometric series and $\sum (1/n)$ is the divergent harmonic series, the given series diverges by Theorem (11.20). ■

We shall conclude this section by returning to the discussion of the epidemic given at the end of Section 11.1. Suppose that instead of I_n (the number ill on day n) we are interested in the total number S_n of individuals who have been ill at some time between the first and n th days. As in our earlier discussion, let us overestimate S_n by approximating the number F_{n+1} of new cases on day $n + 1$ by aNI_n . Thus

$$\begin{aligned} S_n &= I_1 + F_2 + F_3 + F_4 + \cdots + F_n \\ &= I_1 + aNI_1 + aNI_2 + aNI_3 + \cdots + aNI_{n-1}. \end{aligned}$$

Recalling that $I_n = r^{n-1}I_1$, where $r = 1 + aN - b$, we obtain

$$\begin{aligned} S_n &= I_1 + aNI_1 + aNrI_1 + aNr^2I_1 + \cdots + aNr^{n-2}I_1 \\ &= I_1 + aNI_1(1 + r + r^2 + \cdots + r^{n-2}). \end{aligned}$$

As in the proof of Theorem (11.13), this may be written

$$S_n = I_1 + aNI_1 \left(\frac{1}{1-r} - \frac{r^{n-1}}{1-r} \right).$$

If $r < 1$, then

$$\begin{aligned} \lim_{n \rightarrow \infty} S_n &= I_1 + aNI_1 \left(\frac{1}{1-r} \right) \\ &= I_1 \left(1 + \frac{aN}{1-r} \right) \\ &= I_1 \left(1 + \frac{aN}{b - aN} \right) \\ &= I_1 \left(\frac{b}{b - aN} \right). \end{aligned}$$

If a and b are approximated from early data, this result enables health officials to determine an upper bound for the total number of individuals who will be ill at some stage of the epidemic.

11.2 Exercises

In Exercises 1–20 determine whether the infinite series converges or diverges, and if it converges find its sum.

1 $3 + \frac{3}{4} + \cdots + \frac{3}{4^{n-1}} + \cdots$

2 $3 + \frac{3}{(-4)} + \cdots + \frac{3}{(-4)^{n-1}} + \cdots$

3 $1 + \frac{(-1)}{\sqrt{5}} + \cdots + \left(\frac{-1}{\sqrt{5}} \right)^{n-1} + \cdots$

4 $1 + \left(\frac{e}{3} \right) + \cdots + \left(\frac{e}{3} \right)^{n-1} + \cdots$

5 $0.37 + 0.0037 + \cdots + \frac{37}{(100)^n} + \cdots$

6 $0.628 + 0.000628 + \cdots + \frac{628}{(1000)^n} + \cdots$

7 $\sum_{n=1}^{\infty} 2^{-n} 3^{n-1}$

8 $\sum_{n=1}^{\infty} (-5)^{n-1} 4^{-n}$

9 $\sum_{n=1}^{\infty} (-1)^{n-1}$

10 $\sum (\sqrt{2})^{n-1}$

11 $\frac{1}{4 \cdot 5} + \frac{1}{5 \cdot 6} + \cdots + \frac{1}{(n+3)(n+4)} + \cdots$

12 $\frac{-1}{1 \cdot 2} + \frac{-1}{2 \cdot 3} + \cdots + \frac{-1}{n(n+1)} + \cdots$

13 $\frac{5}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \cdots + \frac{5}{n(n+1)} + \cdots$

14 $\frac{1}{4} + \frac{1}{5} + \cdots + \frac{1}{n+3} + \cdots$

15 $3 + \frac{3}{2} + \cdots + \frac{3}{n} + \cdots$

16 $\frac{1}{2} + \frac{2}{3} + \cdots + \frac{n}{n+1} + \cdots$

17 $\sum_{n=1}^{\infty} \frac{3n}{5n-1}$

18 $\sum_{n=1}^{\infty} \frac{1}{1 + (0.3)^n}$

19 $\sum_{n=1}^{\infty} \left(\frac{1}{8^n} + \frac{1}{n(n+1)} \right)$

20 $\sum_{n=1}^{\infty} \left(\frac{1}{3^n} - \frac{1}{4^n} \right)$

Use the examples and theorems discussed in this section to determine whether the series in Exercises 21–28 converge or diverge.

$$\begin{array}{ll}
 21 \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{e}} & 22 \quad \sum_{n=1}^{\infty} \frac{n}{\ln(n+1)} \\
 23 \quad \sum_{n=1}^{\infty} \left(\frac{5}{n+2} - \frac{5}{n+3} \right) & 24 \quad \sum_{n=1}^{\infty} \ln \left(\frac{2n}{7n-5} \right) \\
 25 \quad \sum_{n=1}^{\infty} \left[\left(\frac{3}{2} \right)^n + \left(\frac{2}{3} \right)^n \right] & 26 \quad \sum_{n=1}^{\infty} \left[\frac{1}{n(n+1)} - \frac{4}{n} \right] \\
 27 \quad \sum_{n=1}^{\infty} n \sin \frac{1}{n} & 28 \quad \sum_{n=1}^{\infty} (2^{-n} - 2^{-3n})
 \end{array}$$

Use the method illustrated in Example 1 to prove that the series in Exercises 29 and 30 converge and find their sums.

$$\begin{array}{l}
 29 \quad \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} \\
 30 \quad \sum_{n=1}^{\infty} \frac{-1}{9n^2 + 3n - 2}
 \end{array}$$

- 31 Prove that if $\sum a_n$ diverges, then so does $\sum ca_n$ for every $c \neq 0$.
- 32 Prove Theorem (11.19).
- 33 Prove or disprove: If $\sum a_n$ and $\sum b_n$ both diverge, then $\sum (a_n + b_n)$ diverges.
- 34 What is wrong with the following “proof” that the divergent geometric series $\sum_{n=1}^{\infty} (-1)^{n+1}$ has the sum 0?

$$\begin{aligned}
 \sum_{n=1}^{\infty} (-1)^{n+1} &= [1 + (-1)] + [1 + (-1)] \\
 &\quad + [1 + (-1)] + \cdots \\
 &= 0 + 0 + 0 + \cdots = 0
 \end{aligned}$$

In Exercises 35–38, the bar indicates that the digits underneath it repeat indefinitely. Express each repeating decimal as an infinite series and find the rational number it represents.

$$\begin{array}{ll}
 35 \quad 0.\overline{23} & 36 \quad 5.\overline{146} \\
 37 \quad 3.\overline{2394} & 38 \quad 2.7\overline{1828}
 \end{array}$$

- 39 A rubber ball is dropped from a height of 10 meters. If it rebounds approximately one-half the distance after each

fall, use an infinite geometric series to approximate the total distance the ball travels before coming to rest.

- 40 The bob of a pendulum swings through an arc 24 cm long on its first swing. If each successive swing is approximately five-sixths the length of the preceding swing, use an infinite geometric series to approximate the total distance it travels before coming to rest.
- 41 If a dosage of Q units of a certain drug is administered to an individual, then the amount remaining in the bloodstream at the end of t minutes is given by Qe^{-ct} , where c is a positive constant. Suppose this same dosage is given at successive T -minute intervals.
- Show that the amount $A(k)$ of the drug in the bloodstream immediately after the k th dose is given by $A(k) = \sum_{n=0}^{k-1} Qe^{-ncT}$.
 - Find an upper bound for the amount of the drug in the bloodstream after any number of doses.
 - Find the smallest time between doses that will ensure that $A(k)$ does not exceed a certain level M .
- 42 Suppose that each dollar introduced into the economy recirculates as follows: 85% of the original dollar is spent, then 85% of the remaining \$0.85 is spent, and so on. Find the economic impact (the total amount spent) if \$1,000,000 is introduced into the economy.

Prove that each of the following is true for every real number x in the indicated interval.

- 43 $1 - x + x^2 - x^3 + \cdots + (-1)^{n+1}x^n + \cdots = \frac{1}{1+x}$,
if $-1 < x < 1$
- 44 $1 + x^2 + x^4 + \cdots + x^{2n} + \cdots = \frac{1}{1-x^2}$,
if $-1 < x < 1$
- 45 $\frac{1}{2} - \frac{(x-3)}{4} + \frac{(x-3)^2}{8} - \cdots + \frac{(x-3)^n}{(-2)^{n+1}} + \cdots = \frac{2}{x-1}$,
if $1 < x < 5$
- 46 $3 + (x-1) + \frac{(x-1)^2}{3} + \cdots + \frac{(x-1)^n}{3^{n-1}} + \cdots = \frac{3}{2-x}$,
if $-2 < x < 4$

11.3 Positive Term Series

It is difficult to apply the definition of convergence or divergence (11.12) to an infinite series $\sum a_n$, since in most cases a simple formula for S_n cannot be found. We can, however, develop techniques for using the n th term a_n to test a series for convergence or divergence. When applying these tests we shall be concerned not with the *sum* of the series but with whether the series converges or diverges.

In this section, we shall consider only **positive term series**, that is, series such that $a_n > 0$ for every n .

Theorem (11.21)

If $\sum a_n$ is a positive term series and if there exists a number M such that $S_n < M$ for every n , then the series converges and has a sum $S \leq M$. If no such M exists the series diverges.

Proof If $\{S_n\}$ is the sequence of partial sums of the positive term series $\sum a_n$, then

$$S_1 < S_2 < \cdots < S_n < \cdots$$

and therefore $\{S_n\}$ is monotonic. If there exists a number M such that $S_n < M$ for every n , then as in the proof of Theorem (11.8),

$$\lim_{n \rightarrow \infty} S_n = S \leq M$$

for some S , and hence the series converges. If no such M exists, then $\lim_{n \rightarrow \infty} S_n = \infty$ and the series diverges. $\square$

Suppose a function f is defined for every real number $x \geq 1$. We may then consider the infinite series

$$\sum_{n=1}^{\infty} f(n) = f(1) + f(2) + \cdots + f(n) + \cdots$$

For example, if $f(x) = 1/x^2$, then

$$\sum_{n=1}^{\infty} f(n) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{n^2} + \cdots$$

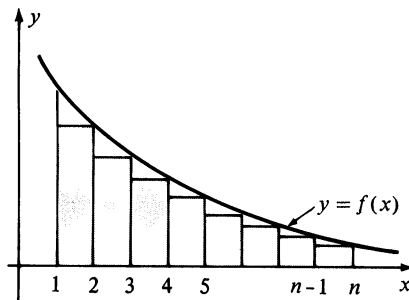
A series of this type may be tested for convergence or divergence by means of an improper integral, as indicated in the following result.

The Integral Test (11.22)

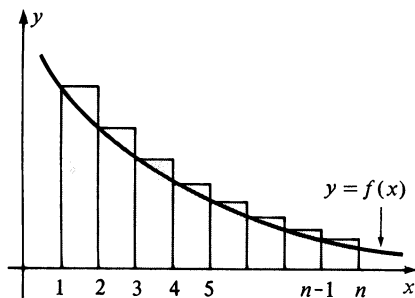
If a function f is positive-valued, continuous, and decreasing for $x \geq 1$, then the infinite series

$$f(1) + f(2) + \cdots + f(n) + \cdots$$

- (i) converges if $\int_1^{\infty} f(x) dx$ converges.
 (ii) diverges if $\int_1^{\infty} f(x) dx$ diverges.



(i)



(ii)

FIGURE 11.3

Proof If n is a positive integer greater than 1, then the area of the inscribed rectangular polygon illustrated in (i) of Figure 11.3 is given by

$$\sum_{k=2}^n f(k) = f(2) + f(3) + \cdots + f(n).$$

Similarly, the area of the circumscribed rectangular polygon illustrated in (ii) of the figure is

$$\sum_{k=1}^{n-1} f(k) = f(1) + f(2) + \cdots + f(n-1).$$

Since $\int_1^n f(x) dx$ is the area under the graph of f from 1 to n ,

$$\sum_{k=2}^n f(k) \leq \int_1^n f(x) dx \leq \sum_{k=1}^{n-1} f(k).$$

If S_n denotes the n th partial sum of the series $f(1) + f(2) + \cdots + f(n) + \cdots$, then this inequality may be written

$$S_n - f(1) \leq \int_1^n f(x) dx \leq S_{n-1}.$$

The preceding inequality implies that if the integral $\int_1^{\infty} f(x) dx$ converges and equals $K > 0$, then

$$S_n - f(1) \leq K, \quad \text{or} \quad S_n \leq K + f(1)$$

for every positive integer n . Hence, by Theorem (11.21) the series $\sum f(n)$ converges. If the improper integral diverges, then

$$\lim_{n \rightarrow \infty} \int_1^n f(x) dx = \infty$$

and since $\int_1^n f(x) dx \leq S_{n-1}$ we also have $\lim_{n \rightarrow \infty} S_{n-1} = \infty$; that is, the series $\sum f(n)$ diverges. $\square$

Example 1 Use the Integral Test to prove that the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

diverges.

Solution If we let $f(x) = 1/x$, then f is positive-valued, continuous, and decreasing for $x \geq 1$ and, therefore, the Integral Test may be applied. Since

$$\begin{aligned} \int_1^{\infty} \frac{1}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\ln x]_1^t \\ &= \lim_{t \rightarrow \infty} [\ln t - \ln 1] = \infty, \end{aligned}$$

the series diverges by (ii) of (11.22). ■

Example 2 Determine whether the infinite series $\sum ne^{-n^2}$ converges or diverges.

Solution If we let $f(x) = xe^{-x^2}$, then the given series is the same as $\sum f(n)$. If $x \geq 1$, f is positive-valued and continuous. The first derivative may be used to determine whether f is decreasing. Since

$$f'(x) = e^{-x^2} - 2x^2e^{-x^2} = e^{-x^2}(1 - 2x^2) < 0,$$

f is decreasing on $[1, \infty)$. We may, therefore, apply the Integral Test as follows:

$$\begin{aligned} \int_1^{\infty} xe^{-x^2} dx &= \lim_{t \rightarrow \infty} \int_1^t xe^{-x^2} dx = \lim_{t \rightarrow \infty} \left(-\frac{1}{2} \right) e^{-x^2} \Big|_1^t \\ &= \left(-\frac{1}{2} \right) \lim_{t \rightarrow \infty} \left[\frac{1}{e^{t^2}} - \frac{1}{e} \right] = \frac{1}{2e}. \end{aligned}$$

Hence the series converges by (i) of (11.22). ■

An integral test may also be used if the function f satisfies the conditions of (11.22) for all $x \geq m$, where m is any positive integer. In this case we merely replace the integral in (11.22) by $\int_m^{\infty} f(x) dx$. This corresponds to deleting the first $m - 1$ terms of the series.

If $f(x) = 1/x^p$ where $p > 0$, then the series $\sum f(n)$ has the form

$$1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

and is called the **p -series**, or **hyperharmonic series**. This series will be very useful when we apply comparison tests later in this section. The following theorem provides information about convergence or divergence.

Theorem (11.23)The p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

converges if $p > 1$ or diverges if $p \leq 1$.

Proof First we note that the special case $p = 1$ is the divergent harmonic series. Next suppose that p is any positive real number different from 1. If we let $f(x) = 1/x^p = x^{-p}$, then f is positive-valued and continuous for $x \geq 1$. Moreover, for these values of x we see that $f'(x) = -px^{-p-1} < 0$, and hence f is decreasing. Thus f satisfies the conditions stated in the Integral Test (11.22) and we consider

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^p} dx &= \lim_{t \rightarrow \infty} \int_1^t x^{-p} dx = \lim_{t \rightarrow \infty} \left[\frac{x^{1-p}}{1-p} \right]_1^t \\ &= \frac{1}{1-p} \lim_{t \rightarrow \infty} [t^{1-p} - 1]. \end{aligned}$$

If $p > 1$, then $p - 1 > 0$ and the last expression may be written

$$\frac{1}{1-p} \lim_{t \rightarrow \infty} \left[\frac{1}{t^{p-1}} - 1 \right] = \frac{1}{1-p} (0 - 1) = \frac{1}{p-1}.$$

Thus by (i) of (11.22), the p -series converges if $p > 1$.

If $0 < p < 1$, then $1 - p > 0$ and

$$\frac{1}{1-p} \lim_{t \rightarrow \infty} [t^{1-p} - 1] = \infty.$$

Hence by (ii) of (11.22), the p -series diverges.

If $p \leq 0$, then $\lim_{n \rightarrow \infty} (1/n^p) \neq 0$, and by (11.15) the series diverges. $\square$

Example 3 Determine whether the following series converge or diverge.

(a) $1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} + \cdots$

(b) $5 + \frac{5}{\sqrt{2}} + \frac{5}{\sqrt{3}} + \cdots + \frac{5}{\sqrt{n}} + \cdots$

Solution

(a) The series $\sum 1/n^2$ converges since it is the p -series with $p = 2 > 1$.

(b) The series $\sum 1/\sqrt{n}$ diverges since it is the p -series with $p = \frac{1}{2} < 1$. Consequently $\sum 5/\sqrt{n}$ diverges. (See Exercise 31 of Section 11.2.) $\blacksquare$

The next theorem allows us to use known convergent (divergent) series to establish the convergence (divergence) of other series.

Basic Comparison Test (11.24)

Suppose $\sum a_n$ and $\sum b_n$ are positive term series.

- (i) If $\sum b_n$ converges and $a_n \leq b_n$ for every positive integer n , then $\sum a_n$ converges.
- (ii) If $\sum b_n$ diverges and $a_n \geq b_n$ for every positive integer n , then $\sum a_n$ diverges.

Proof Let S_n and T_n denote the n th partial sums of $\sum a_n$ and $\sum b_n$, respectively. Suppose $\sum b_n$ converges and has the sum T . If $a_n \leq b_n$ for every n , then $S_n \leq T_n < T$ and hence by Theorem (11.21) $\sum a_n$ converges. This proves part (i).

To prove (ii), suppose $\sum b_n$ diverges and $a_n \geq b_n$ for every n . Then $S_n \geq T_n$ and, since T_n increases without bound as n becomes infinite, so does S_n . Consequently, $\sum a_n$ diverges. $\square$

Since convergence or divergence of a series is not affected by deleting a finite number of terms, the conditions $a_n \geq b_n$ or $a_n \leq b_n$ of (11.24) are only required from the k th term on, where k is some fixed positive integer.

A series $\sum d_n$ is said to **dominate** a series $\sum c_n$ if $0 < c_n \leq d_n$ for every positive integer n . Using this terminology, part (i) of (11.24) states that a positive term series that is dominated by a convergent series is also convergent. Part (ii) states that a series which dominates a divergent series also diverges.

Example 4 Determine whether the following series converge or diverge.

$$(a) \sum_{n=1}^{\infty} \frac{1}{2+5^n} \quad (b) \sum_{n=2}^{\infty} \frac{3}{\sqrt{n}-1}$$

Solution

(a) For every $n \geq 1$,

$$\frac{1}{2+5^n} < \frac{1}{5^n} = \left(\frac{1}{5}\right)^n.$$

Since $\sum (1/5)^n$ is a convergent geometric series, the given series converges, by (i) of (11.24).

(b) The p -series $\sum 1/\sqrt{n}$ diverges and hence so does the series obtained by discarding the first term $1/\sqrt{1}$. Since

$$\frac{3}{\sqrt{n}-1} > \frac{1}{\sqrt{n}} \quad \text{if } n \geq 2,$$

it follows from (ii) of (11.24) that the given series diverges. $\blacksquare$

Limit Comparison Test (11.25)

If $\sum a_n$ and $\sum b_n$ are positive term series and if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = k > 0$$

then either both series converge or both diverge.

Proof If $\lim_{n \rightarrow \infty} (a_n/b_n) = k > 0$, then there exists a number N such that

$$\frac{k}{2} < \frac{a_n}{b_n} < \frac{3k}{2} \quad \text{whenever } n > N.$$

This is equivalent to

$$\frac{k}{2} b_n < a_n < \frac{3k}{2} b_n \quad \text{whenever } n > N.$$

If the series $\sum a_n$ converges, then $\sum (k/2)b_n$ also converges, since it is dominated by $\sum a_n$. Applying (ii) of (11.19),

$$\sum b_n = \sum \left(\frac{2}{k}\right) \left(\frac{k}{2}\right) b_n$$

converges. Conversely, if $\sum b_n$ converges, then so does $\sum a_n$, since it is dominated by the convergent series $\sum (3k/2)b_n$. We have proved that $\sum a_n$ converges if and only if $\sum b_n$ converges. Consequently, $\sum a_n$ diverges if and only if $\sum b_n$ diverges. $\square$

Two other comparison tests are stated in Exercises 49 and 50.

Example 5 Determine whether the following series converge or diverge.

$$(a) \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2 + 1}} \quad (b) \sum_{n=1}^{\infty} \frac{3n^2 + 5n}{2^n(n^2 + 1)}$$

Solution

(a) We shall apply (11.25) with

$$a_n = \frac{1}{\sqrt[3]{n^2 + 1}} \quad \text{and} \quad b_n = \frac{1}{\sqrt[3]{n^2}} = \frac{1}{n^{2/3}}.$$

The series $\sum b_n$ is the p -series with $p = \frac{2}{3} < 1$ and, therefore, diverges. Since

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^2}}{\sqrt[3]{n^2 + 1}} = \lim_{n \rightarrow \infty} \sqrt[3]{\frac{n^2}{n^2 + 1}} = 1 > 0,$$

$\sum a_n$ also diverges.

(b) If we let

$$a_n = \frac{3n^2 + 5n}{2^n(n^2 + 1)} \quad \text{and} \quad b_n = \frac{1}{2^n}$$

then
$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{3n^2 + 5n}{n^2 + 1} = 3 > 0.$$

Since $\sum b_n$ is a convergent geometric series, the series $\sum a_n$ is also convergent by (11.25). ■

When seeking a suitable series $\sum b_n$ to be used for comparison with $\sum a_n$, where a_n is a fraction, a good procedure is to discard all terms in the numerator and denominator except those that have the most effect on the magnitude. For example, if we employ this technique in part (b) of Example 5, where

$$a_n = \frac{3n^2 + 5n}{2^n n^2 + 2^n}$$

we are led to
$$\frac{3n^2}{2^n n^2} = \frac{3}{2^n}.$$

This suggests taking $b_n = 1/2^n$, as was done in the solution.

Example 6 If
$$a_n = \frac{8n + \sqrt{n}}{5 + n^2 + n^{7/2}}$$

determine whether $\sum a_n$ converges or diverges.

Solution To find a suitable comparison series $\sum b_n$ we discard all but the highest powers of n in the numerator and denominator, obtaining

$$\frac{8n}{n^{7/2}} = \frac{8}{n^{5/2}}.$$

This suggests that we choose $b_n = 1/n^{5/2}$. In this event

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{8n^{7/2} + n^3}{5 + n^2 + n^{7/2}} = 8 > 0.$$

Since $\sum b_n$ is a convergent p -series with $p = \frac{5}{2} > 1$, it follows from (11.25) that $\sum a_n$ is also convergent. ■

We shall conclude this section with several general remarks about positive term series. Suppose $\sum a_n$ is a positive term series and the terms are grouped in some manner. For example, we could have

$$(a_1 + a_2) + a_3 + (a_4 + a_5 + a_6 + a_7) + \cdots$$

If we denote the last series by $\sum b_n$, so that

$$b_1 = a_1 + a_2, \quad b_2 = a_3, \quad b_3 = a_4 + a_5 + a_6 + a_7, \quad \dots$$

then any partial sum of the series $\sum b_n$ is also a partial sum of $\sum a_n$. It follows that if $\sum a_n$ converges, then $\sum b_n$ converges and has the same sum. A similar argument may be used for any grouping of the terms of $\sum a_n$. Thus, *if a positive term series converges, then the series obtained by grouping the terms in any manner also converges and has the same sum.*

This result is not necessarily true for a series that contains both positive and negative terms. Also, we cannot make a similar statement about arbitrary divergent series. For example, the terms of the divergent series $\sum (-1)^n$ may be grouped to produce a convergent series (see Exercise 34 of Section 11.2).

Next, suppose that a convergent positive term series $\sum a_n$ has the sum S and that a new series $\sum b_n$ is formed by rearranging the terms in some way. For example, $\sum b_n$ could be the series

$$a_2 + a_8 + a_1 + a_5 + a_7 + a_3 + \dots$$

If T_n is the n th partial sum of $\sum b_n$, then it is a sum of terms of $\sum a_n$. If m is the largest subscript associated with the terms a_i in T_n , then $T_n \leq S_m < S$. Consequently, $T_n < S$ for every n . Applying Theorem (11.21), $\sum b_n$ converges and has a sum $T \leq S$. The preceding proof is independent of the particular rearrangement of terms. We may also regard the series $\sum a_n$ as having been obtained by rearranging the terms of $\sum b_n$ and hence, by the same argument, $S \leq T$. We have proved that *if the terms of a convergent positive term series $\sum a_n$ are rearranged in any manner, then the resulting series converges and has the same sum.*

11.3 Exercises

Use the Integral Test to determine whether the series in Exercises 1–16 converge or diverge.

1 $\sum_{n=1}^{\infty} \frac{1}{(3+2n)^2}$

2 $\sum_{n=1}^{\infty} \frac{1}{(4+n)^{3/2}}$

3 $\sum_{n=1}^{\infty} \frac{1}{4n+7}$

4 $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

5 $\sum_{n=1}^{\infty} \frac{\ln n}{n}$

6 $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$

7 $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{2n+1}}$

8 $\sum_{n=1}^{\infty} \frac{1}{1+16n^2}$

9 $\sum_{n=1}^{\infty} \frac{\arctan n}{1+n^2}$

10 $\sum_{n=1}^{\infty} ne^{-n}$

11 $\sum_{n=3}^{\infty} \frac{1}{n(2n-5)}$

12 $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

13 $\sum_{n=1}^{\infty} n2^{-n^2}$

15 $\sum_{n=2}^{\infty} \frac{1}{n\sqrt[3]{\ln n}}$

14 $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+9}}$

16 $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$

Use comparison tests to determine whether the series in Exercises 17–38 converge or diverge.

17 $\sum_{n=1}^{\infty} \frac{1}{n^4+n^2+1}$

18 $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$

19 $\sum_{n=1}^{\infty} \frac{1}{n3^n}$

20 $\sum_{n=1}^{\infty} \frac{n^2}{n^3+1}$

21 $\sum_{n=1}^{\infty} \frac{2n+n^2}{n^3+1}$

22 $\sum_{n=1}^{\infty} \frac{2}{3+\sqrt{n}}$

23 $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n+4}$

24 $\sum_{n=1}^{\infty} \frac{n^5+4n^3+1}{2n^8+n^4+2}$

$$25 \quad \sum_{n=2}^{\infty} \frac{1}{\sqrt{4n^3 - 5n}}$$

$$27 \quad \sum_{n=1}^{\infty} \frac{8n^2 - 7}{e^n(n+1)^2}$$

$$29 \quad \sum_{n=1}^{\infty} \frac{1 + 2^n}{1 + 3^n}$$

$$31 \quad \sum_{n=1}^{\infty} \frac{2 + \cos n}{n^2}$$

$$33 \quad \sum_{n=1}^{\infty} \frac{(2n+1)^3}{(n^3+1)^2}$$

$$35 \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{5n^2+1}}$$

$$37 \quad \sum_{n=1}^{\infty} \frac{1}{n^n}$$

$$26 \quad \sum_{n=4}^{\infty} \frac{3n}{2n^2 - 7}$$

$$28 \quad \sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n}$$

$$30 \quad \sum_{n=1}^{\infty} \frac{\ln n}{n^4} \text{ (Hint: } \ln n < n \text{)}$$

$$32 \quad \sum_{n=1}^{\infty} \frac{\arctan n}{n^2}$$

$$34 \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)(n+2)}}$$

$$36 \quad \sum_{n=1}^{\infty} \frac{3n+5}{n \cdot 2^n}$$

$$38 \quad \sum_{n=1}^{\infty} \frac{1}{n!}$$

Determine whether the series in Exercises 39–46 converge or diverge.

$$39 \quad \sum_{n=1}^{\infty} \frac{n + \ln n}{n^3 + n + 1}$$

$$40 \quad \sum_{n=1}^{\infty} \frac{n + \ln n}{n^2 + 1}$$

$$41 \quad \sum_{n=1}^{\infty} \sin \frac{1}{n^2}$$

$$42 \quad \sum_{n=1}^{\infty} \tan \frac{1}{n}$$

$$43 \quad \sum_{n=1}^{\infty} \frac{\ln n}{n^3}$$

$$44 \quad \sum_{n=1}^{\infty} \ln \left(1 + \frac{1}{2^n} \right)$$

$$45 \quad \sum_{n=1}^{\infty} \frac{n^2 + 2^n}{n + 3^n}$$

$$46 \quad \sum_{n=1}^{\infty} \frac{\sin n + 2^n}{n + 5^n}$$

In Exercises 47 and 48 find all real numbers k for which the series converge.

$$47 \quad \sum_{n=2}^{\infty} \frac{1}{n^k \ln n}$$

$$48 \quad \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^k}$$

49 Suppose $\sum a_n$ and $\sum b_n$ are positive term series. Prove that if $\lim_{n \rightarrow \infty} (a_n/b_n) = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges. (This is not necessarily true for series that contain negative terms.)

50 Prove that if $\lim_{n \rightarrow \infty} (a_n/b_n) = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

51 Let $\sum a_n$ be a convergent positive series. Let $f(x) = a_x$, and suppose f is continuous and decreasing for $x \geq N$, where N is an integer. Prove that the error in approximating the given series by $\sum_{n=1}^N a_n$ is less than $\int_N^{\infty} f(x) dx$.

Use Exercise 51 to determine the smallest number of terms that can be added to approximate the sum of the series in Exercises 52–54 with an error less than E .

$$52 \quad \sum_{n=1}^{\infty} \frac{1}{n^2}, E = 0.001$$

$$53 \quad \sum_{n=1}^{\infty} \frac{1}{n^3}, E = 0.01$$

$$54 \quad \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}, E = 0.05$$

55 Prove that if a positive term series $\sum a_n$ converges, then $\sum (1/a_n)$ diverges.

56 Prove that if a positive terms series $\sum a_n$ converges, then $\sum \sqrt{a_n a_{n+1}}$ converges.

(Hint: First show that $\sqrt{a_n a_{n+1}} \leq (a_n + a_{n+1})/2$.)

11.4 Alternating Series

An infinite series whose terms are alternately positive and negative is called an **alternating series**. It is customary to express an alternating series in one of the forms

$$a_1 - a_2 + a_3 - a_4 + \cdots + (-1)^{n-1} a_n + \cdots$$

or

$$-a_1 + a_2 - a_3 + a_4 - \cdots + (-1)^n a_n + \cdots$$

where each $a_i > 0$. The next theorem provides the main test for convergence of these series.

Alternating Series Test (11.26)

If $a_k \geq a_{k+1} > 0$ for every positive integer k and $\lim_{n \rightarrow \infty} a_n = 0$, then the alternating series $\sum (-1)^{n-1} a_n$ is convergent.

Proof Let us first consider the partial sums

$$S_2, S_4, S_6, \dots, S_{2n}, \dots$$

that contain an even number of terms of the series. Since

$$S_{2n} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n})$$

and $a_k - a_{k+1} \geq 0$ for every k , we see that

$$0 \leq S_2 \leq S_4 \leq \cdots \leq S_{2n} \leq \cdots,$$

that is, $\{S_{2n}\}$ is a monotonic sequence. The formula for S_{2n} can also be written

$$S_{2n} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n}$$

and hence $S_{2n} \leq a_1$ for every positive integer n . As in the proof of Theorem (11.8),

$$\lim_{n \rightarrow \infty} S_{2n} = S \leq a_1$$

for some number S . If we consider a partial sum S_{2n+1} having an *odd* number of terms of the series, then $S_{2n+1} = S_{2n} + a_{2n+1}$ and, since $\lim_{n \rightarrow \infty} a_{2n+1} = 0$,

$$\lim_{n \rightarrow \infty} S_{2n+1} = \lim_{n \rightarrow \infty} S_{2n} = S.$$

It follows that

$$\lim_{n \rightarrow \infty} S_n = S \leq a_1$$

that is, the series converges. □

Example 1 Determine whether the following alternating series converge or diverge.

$$(a) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2n}{4n^2 - 3} \quad (b) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2n}{4n - 3}$$

Solution

$$(a) \text{ Let } a_n = f(n) = \frac{2n}{4n^2 - 3}.$$

To apply the Alternating Series Test we must show that:

- (i) $a_k \geq a_{k+1}$ for every positive integer k .
- (ii) $\lim_{n \rightarrow \infty} a_n = 0$.

One method of proving (i) is to show that $f(x) = 2x/(4x^2 - 3)$ is decreasing for $x \geq 1$. By the Quotient Rule,

$$\begin{aligned} f'(x) &= \frac{(4x^2 - 3)(2) - (2x)(8x)}{(4x^2 - 3)^2} \\ &= \frac{-8x^2 - 6}{(4x^2 - 3)^2} < 0. \end{aligned}$$

It follows that f is decreasing and, therefore, $a_k \geq a_{k+1}$ for every positive integer k .

We can obtain the same result directly by proving that $a_k - a_{k+1} \geq 0$. Specifically, if $a_n = 2n/(4n^2 - 3)$, then

$$\begin{aligned} a_k - a_{k+1} &= \frac{2k}{4k^2 - 3} - \frac{2(k+1)}{4(k+1)^2 - 3} \\ &= \frac{8k^2 + 8k + 6}{(4k^2 - 3)(4k^2 + 8k + 1)} \geq 0 \end{aligned}$$

for every positive integer k . Another technique for proving that $a_k \geq a_{k+1}$ is to show that $a_{k+1}/a_k \leq 1$.

To prove (ii) we see that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2n}{4n^2 - 3} = 0.$$

Thus the alternating series converges.

(b) It can be shown that $a_k \geq a_{k+1}$ for every k ; however,

$$\lim_{n \rightarrow \infty} \frac{2n}{4n - 3} = \frac{1}{2} \neq 0$$

and hence the series diverges by (11.15). ■

If an infinite series converges, then the n th partial sum S_n can be used to approximate the sum S of the series. In most cases it is difficult to determine the accuracy of the approximation. However, for an *alternating* series, the next theorem provides a simple way of estimating the error which is involved.

Theorem (11.27)

If $\sum (-1)^{n-1} a_n$ is an alternating series such that $a_k > a_{k+1} > 0$ for every positive integer k and if $\lim_{n \rightarrow \infty} a_n = 0$, then the error involved in approximating the sum S of the series by the n th partial sum S_n is numerically less than a_{n+1} .

Proof Note that the alternating series $\sum (-1)^{n-1} a_n$ satisfies the conditions of the Alternating Series Test and hence has a sum S . The series obtained by deleting the first n terms, namely

$$(-1)^n a_{n+1} + (-1)^{n+1} a_{n+2} + (-1)^{n+2} a_{n+3} + \cdots$$

also satisfies the conditions of (11.26) and, therefore, has a sum R_n . Thus

$$S - S_n = R_n = (-1)^n(a_{n+1} - a_{n+2} + a_{n+3} - \cdots)$$

and
$$|R_n| = a_{n+1} - a_{n+2} + a_{n+3} - \cdots$$

Employing the same argument used in the proof of the Alternating Series Test (but replacing $\leq$ by $<$) we see that $|R_n| < a_{n+1}$. Consequently,

$$|S - S_n| = |R_n| < a_{n+1}$$

which is what we wished to prove. $\square$

Example 2 Prove that the series

$$1 - \frac{1}{3!} + \frac{1}{5!} - \cdots + (-1)^{n-1} \frac{1}{(2n-1)!} + \cdots$$

is convergent and approximate its sum S to five decimal places.

Solution Evidently $a_n = 1/(2n-1)!$ has the limit 0 as $n \rightarrow \infty$, and $a_k > a_{k+1}$ for every positive integer k . Hence the series converges by the Alternating Series Test. If we use S_4 to approximate the sum S of the series, then

$$\begin{aligned} S &\approx 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} \\ &= 1 - \frac{1}{6} + \frac{1}{120} - \frac{1}{5040} \approx 0.84147. \end{aligned}$$

By Theorem (11.27) the error involved in the approximation is less than

$$a_5 = \frac{1}{9!} < 0.000005.$$

Thus the approximation 0.84147 is accurate to five decimal places. It will follow from (11.42) that the sum of the given series equals $\sin 1$ and hence $\sin 1 \approx 0.84147$. This illustrates one technique for constructing trigonometric tables. $\blacksquare$

11.4 Exercises

Determine whether the series in Exercises 1–14 converge or diverge.

1 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\sqrt{2n+1}}$

2 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^{2/3}}$

5 $\sum_{n=2}^{\infty} (-1)^n \frac{n}{\ln n}$

6 $\sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$

7 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^2+1}{n^3+1}$

8 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{3n+4}{5n+7}$

3 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\ln(n+1)}$

4 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+4}$

9 $\sum_{n=1}^{\infty} (-1)^n \frac{e^n}{n^4}$

10 $\sum_{n=0}^{\infty} (-1)^n \frac{\sqrt{n+1}}{8n+5}$

$$11 \quad \sum_{n=1}^{\infty} (-1)^n \frac{\sqrt[3]{n}}{2n+5}$$

$$12 \quad \sum_{n=0}^{\infty} (-1)^n \frac{1+4^n}{1+3^n}$$

$$19 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n+1}{5^n}$$

$$20 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{n} \left(\frac{1}{2}\right)^n$$

$$13 \quad \sum_{n=1}^{\infty} (-1)^n \frac{(\ln n)^k}{n}, \text{ where } k \text{ is any positive integer.}$$

$$14 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt[k]{n}}, \text{ where } k \text{ is any positive real number.}$$

In each of Exercises 21–24, use Theorem (11.27) to find a positive integer n such that S_n approximates the sum of the given series to four decimal places.

$$21 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$$

$$22 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n}}$$

$$23 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{n^n}$$

$$24 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{n^3 + 1}$$

$$15 \quad \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!}$$

$$16 \quad \sum_{n=0}^{\infty} (-1)^{n+1} \frac{1}{(2n)!}$$

25 If $\sum a_n$ and $\sum b_n$ are both convergent series, is $\sum a_n b_n$ convergent? Explain.

$$17 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^3}$$

$$18 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^5}$$

26 If $\sum a_n$ and $\sum b_n$ are both divergent series, is $\sum a_n b_n$ divergent? Explain.

In Exercises 15–20, approximate the sum of each series to three decimal places.

11.5 Absolute Convergence

The following concept plays an important role in work with infinite series.

Definition (11.28)

An infinite series $\sum a_n$ is **absolutely convergent** if the series

$$\sum |a_n| = |a_1| + |a_2| + \cdots + |a_n| + \cdots$$

obtained by taking the absolute value of each term is convergent.

Note that if $\sum a_n$ is a positive term series, then $|a_n| = a_n$, and in this case absolute convergence is the same as convergence.

Example 1 Prove that the following alternating series is absolutely convergent:

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots + (-1)^n \frac{1}{n^2} + \cdots$$

Solution Taking the absolute value of each term gives us

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots + \frac{1}{n^2} + \cdots$$

which is a convergent p -series. Hence by Definition (11.28), the alternating series is absolutely convergent. ■

The next theorem tells us that absolute convergence implies convergence.

Theorem (11.29)

If an infinite series $\sum a_n$ is absolutely convergent, then $\sum a_n$ is convergent.

Proof
then

If we let $b_n = a_n + |a_n|$ and use the property $-|a_n| \leq a_n \leq |a_n|$,

$$0 \leq a_n + |a_n| \leq 2|a_n|, \quad \text{or} \quad 0 \leq b_n \leq 2|a_n|.$$

If $\sum a_n$ is absolutely convergent, then $\sum |a_n|$ is convergent, and hence by (ii) of Theorem (11.19), $\sum 2|a_n|$ is convergent. Applying the Comparison Test (11.24) it follows that $\sum b_n$ is convergent. Using (iii) of (11.19), $\sum (b_n - |a_n|)$ is convergent. Since $b_n - |a_n| = a_n$, this completes the proof. $\square$

Example 2 Determine whether the following series is convergent or divergent:

$$\sin 1 + \frac{\sin 2}{2^2} + \frac{\sin 3}{3^2} + \cdots + \frac{\sin n}{n^2} + \cdots$$

Solution The series contains both positive and negative terms, but it is not an alternating series since, for example, the first three terms are positive and the next three are negative. The series of absolute values is

$$\sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{|\sin n|}{n^2}.$$

Since
$$\frac{|\sin n|}{n^2} \leq \frac{1}{n^2}$$

the series of absolute values $\sum |(\sin n)/n^2|$ is dominated by the convergent p -series $\sum 1/n^2$ and hence is convergent. Thus the given series is absolutely convergent and, therefore, is convergent by Theorem (11.29). $\blacksquare$

Series that are convergent, but are not absolutely convergent, are given a special name, as indicated in the next definition.

Definition (11.30)

An infinite series $\sum a_n$ is **conditionally convergent** if $\sum a_n$ is convergent and $\sum |a_n|$ is divergent.

Example 3 Show that the following alternating series is conditionally convergent:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + (-1)^{n-1} \frac{1}{n} + \cdots$$

Solution The series is convergent by the Alternating Series Test. Taking the absolute value of each term, we obtain

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} + \cdots$$

which is the divergent harmonic series. Hence by Definition (11.30), the alternating series is conditionally convergent. ■

We see from the preceding discussion that an arbitrary infinite series may be classified in exactly *one* of the following ways: (i) absolutely convergent, (ii) conditionally convergent, (iii) divergent. Of course, for positive term series we need only determine convergence or divergence.

One of the most important tests for absolute convergence is the following.

The Ratio Test (11.31)

Let $\sum a_n$ be an infinite series of nonzero terms.

- (i) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, the series is absolutely convergent.
- (ii) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$, or $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$, the series is divergent.
- (iii) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, the series may be absolutely convergent, conditionally convergent, or divergent.

Proof (i) Suppose $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = L < 1$. If r is any number such that $0 \leq L < r < 1$, then there exists an integer N such that

$$\left| \frac{a_{n+1}}{a_n} \right| < r \quad \text{whenever } n \geq N.$$

This implies that

$$\begin{aligned} |a_{N+1}| &< |a_N| r \\ |a_{N+2}| &< |a_{N+1}| r < |a_N| r^2 \\ |a_{N+3}| &< |a_{N+2}| r < |a_N| r^3 \end{aligned}$$

and, in general,

$$|a_{N+m}| < |a_N| r^m \quad \text{whenever } m > 0.$$

It follows from the Comparison Test (11.24) that the series

$$|a_{N+1}| + |a_{N+2}| + \cdots + |a_{N+m}| + \cdots$$

converges, since its terms are less than the corresponding terms of the convergent geometric series

$$|a_N|r + |a_N|r^2 + \cdots + |a_N|r^n + \cdots$$

Since convergence or divergence is unaffected by discarding a finite number of terms, the series $\sum_{n=1}^{\infty} |a_n|$ also converges, that is, $\sum a_n$ is absolutely convergent.

(ii) Suppose next that $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = L > 1$. If r is a number such that $1 < r < L$, then there exists an integer N such that

$$\left| \frac{a_{n+1}}{a_n} \right| > r > 1 \quad \text{whenever } n \geq N.$$

Consequently, $|a_{n+1}| > |a_n|$ if $n \geq N$ and therefore $\lim_{n \rightarrow \infty} |a_n| \neq 0$. Thus $\lim_{n \rightarrow \infty} a_n \neq 0$ and, by (11.15), the series $\sum a_n$ diverges. The proof for $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = \infty$ is similar and is left as an exercise.

(iii) The Ratio Test provides no useful information if

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1.$$

Indeed, it is easy to verify that the limit is 1 for the absolutely convergent series $\sum (-1)^n/n^2$, for the conditionally convergent series $\sum (-1)^n/n$, and for the divergent series $\sum 1/n$. Consequently, *if the limit is 1, then a different test should be employed.* $\square$

Example 4 Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.

$$(a) \sum_{n=1}^{\infty} (-1)^n \frac{3^n}{n!} \qquad (b) \sum_{n=1}^{\infty} \frac{3^n}{n^2}$$

Solution

(a) Since

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{3}{n+1} = 0 < 1 \end{aligned}$$

the series is absolutely convergent by the Ratio Test.

(b) Since the series contains only positive terms the absolute value signs in Theorem (11.31) may be deleted. Thus

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \frac{3^{n+1}}{(n+1)^2} \cdot \frac{n^2}{3^n} \\ &= \lim_{n \rightarrow \infty} \frac{3n^2}{n^2 + 2n + 1} = 3 > 1\end{aligned}$$

and hence the series diverges by the Ratio Test. ■

Example 5 Determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{n^n}{n!}$.

Solution Since all terms of the series are positive, we may drop the absolute value signs in the Ratio Test (11.31). Thus

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1)} \cdot \frac{1}{n^n} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e\end{aligned}$$

where the last equality is a consequence of Theorem (7.27). Since $e > 1$, the series diverges by the Ratio Test. ■

The following test is useful if a_n contains only powers of n . It is not as versatile as the Ratio Test because it cannot be applied if a_n contains factorials.

The Root Test (11.32)

Let $\sum a_n$ be an infinite series.

- (i) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L < 1$, the series is absolutely convergent.
- (ii) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L > 1$, or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$, the series is divergent.
- (iii) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$, the series may be absolutely convergent, conditionally convergent, or divergent.

Proof The proof is similar to that used for the Ratio Test. If $L < 1$ as in (i), let us consider any number r such that $L < r < 1$. By the definition of limit, there exists a positive integer N such that if $n \geq N$,

$$\sqrt[n]{|a_n|} < r, \quad \text{or} \quad |a_n| < r^n.$$

Since $0 < r < 1$, $\sum_{n=N}^{\infty} r^n$ is a convergent geometric series, and hence by the Comparison Test (11.24), $\sum_{n=N}^{\infty} |a_n|$ converges. Consequently, $\sum_{n=1}^{\infty} |a_n|$

converges; that is, $\sum a_n$ is absolutely convergent. This proves (i). The remainder of the proof is left as an exercise. $\square$

Example 6 Determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{2^{3n+1}}{n^n}$.

Solution Since all terms are positive we may delete the absolute value signs in the Root Test (11.32). Thus

$$\begin{aligned}\lim_{n \rightarrow \infty} \sqrt[n]{\frac{2^{3n+1}}{n^n}} &= \lim_{n \rightarrow \infty} \left(\frac{2^{3n+1}}{n^n} \right)^{1/n} \\ &= \lim_{n \rightarrow \infty} \frac{2^{3+(1/n)}}{n} = 0 < 1.\end{aligned}$$

Hence the series converges by the Root Test. $\blacksquare$

It can be proved that if a series $\sum a_n$ is absolutely convergent and if the terms are rearranged in any manner, then the resulting series converges and has the same sum as the given series. This is not true for conditionally convergent series. Indeed, if $\sum a_n$ is conditionally convergent, then by suitably rearranging terms one can obtain either a divergent series, or a series that converges and has any desired sum S .*

We now have at our disposal a variety of tests that can be used to investigate an infinite series for convergence or divergence. It takes considerable skill to determine which test is best suited for a particular series. This skill can only be obtained by working many exercises involving different types of series. The following guidelines may be helpful in deciding which test to apply. There are, however, infinite series that are not readily handled by the guidelines, even by experts. In those cases it may be necessary to use methods developed in advanced mathematics courses.

Guidelines for investigating an infinite series $\sum a_n$

- I** If $\lim_{n \rightarrow \infty} a_n \neq 0$, the series diverges.
- II** If $\lim_{n \rightarrow \infty} a_n = 0$, proceed as follows.
 - A** If $\sum a_n$ is a positive term series, use one of the following tests.
 - (1) Basic Comparison Test
 - (2) Limit Comparison Test
 - (3) Integral Test
 - (4) Ratio Test
 - (5) Root Test

* See, for example, R. C. Buck, *Advanced Calculus*, Third Edition (New York: McGraw-Hill, 1978), pp. 238–239.

Remarks: When applying (1) or (2), consider a suitable p -series or geometric series. Of the two comparison tests, (1) is seldom used to investigate a series, since it is usually more difficult to apply than (2).

Use (3) if $f(x) = a_x$ is easy to integrate.

Use (4) or (5) if factorials, or exponents involving n , occur in a_n .

B If $\sum a_n$ is an alternating series, either

(1) use the Alternating Series Test, or

(2) apply a test from Part A to $\sum |a_n|$. If $\sum |a_n|$ is convergent, then so is $\sum a_n$.

C If $\sum a_n$ is neither a positive term series nor an alternating series, apply a test from Part A to $\sum |a_n|$. If $\sum |a_n|$ is convergent, then so is $\sum a_n$.

11.5 Exercises

In Exercises 1–38, determine whether (a) a series which contains both positive and negative terms is absolutely convergent, conditionally convergent, or divergent; or (b) a positive term series is convergent or divergent.

$$1 \quad \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}}$$

$$2 \quad \sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2 + 1}$$

$$3 \quad \sum_{n=2}^{\infty} (-1)^n \frac{5}{n^4 - 1}$$

$$4 \quad \sum_{n=1}^{\infty} (-1)^n e^{-n}$$

$$5 \quad \sum_{n=1}^{\infty} \frac{1000 - n}{n!}$$

$$6 \quad \sum_{n=1}^{\infty} \frac{n!}{e^n}$$

$$7 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{3n + 1}{2^n}$$

$$8 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{3^n}{n^2 + 4}$$

$$9 \quad \sum_{n=1}^{\infty} \frac{5^n}{n(3^{n+1})}$$

$$10 \quad \sum_{n=1}^{\infty} \frac{2^{n-1}}{5^n(n+1)}$$

$$11 \quad \sum_{n=1}^{\infty} \frac{(-100)^n}{n!}$$

$$12 \quad \sum_{n=1}^{\infty} \frac{10 - n^2}{n!}$$

$$13 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n^2 + 1}$$

$$14 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n + 1}$$

$$15 \quad \sum_{n=1}^{\infty} (-1)^n \frac{n^2 + 1}{n^3 + 1}$$

$$16 \quad \sum_{n=1}^{\infty} (-1)^n \frac{2n + 1}{n^2 + n^3}$$

$$17 \quad \sum_{n=2}^{\infty} (-1)^n \frac{1}{n(\ln n)^5}$$

$$18 \quad \sum_{n=1}^{\infty} (-1)^n \frac{n!}{(n+1)^5}$$

$$19 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2}{n^3 + e^n}$$

$$20 \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n3^{2n}}{5^{n-1}}$$

$$21 \quad \sum_{n=1}^{\infty} (-1)^n \frac{\arctan n}{n^2}$$

$$22 \quad \sum_{n=1}^{\infty} \frac{\cos n - 1}{n^{3/2}}$$

$$23 \quad \sum_{n=1}^{\infty} \frac{n^n}{10^n}$$

$$24 \quad \sum_{n=1}^{\infty} \frac{10 - 2^n}{n!}$$

$$25 \quad \sum_{n=1}^{\infty} \frac{n!}{(-5)^n}$$

$$26 \quad \sum_{n=1}^{\infty} \frac{\sec^{-1} n}{\tan^{-1} n}$$

$$27 \quad \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$$

$$28 \quad \sum_{n=1}^{\infty} (-1)^n \frac{n^2 + 3}{(2n - 5)^2}$$

$$29 \quad \sum_{n=1}^{\infty} \frac{\sin \sqrt{n}}{\sqrt{n^3 + 1}}$$

$$30 \quad \sum_{n=1}^{\infty} \frac{n!}{n^n}$$

$$31 \quad 1 + \frac{1 \cdot 3}{2!} + \frac{1 \cdot 3 \cdot 5}{3!} + \cdots + \frac{1 \cdot 3 \cdot 5 \cdots (2n - 1)}{n!} + \cdots$$

$$32 \quad \frac{1}{2} + \frac{1 \cdot 4}{2 \cdot 4} + \frac{1 \cdot 4 \cdot 7}{2 \cdot 4 \cdot 6} + \cdots + \frac{1 \cdot 4 \cdot 7 \cdots (3n - 2)}{2 \cdot 4 \cdot 6 \cdots (2n)} + \cdots$$

$$33 \quad \sum_{n=2}^{\infty} \frac{1}{(\ln n)^n}$$

$$34 \quad \sum_{n=1}^{\infty} \frac{(2n)^n}{(5n + 3n^{-1})^n}$$

$$35 \quad \sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{(1.01)^n}$$

$$36 \quad \sum_{n=1}^{\infty} (-1)^n 3^{1/n}$$

$$37 \quad \sum_{n=1}^{\infty} (-1)^n n \tan \frac{1}{n}$$

$$38 \quad \sum_{n=1}^{\infty} \frac{\cos(n\pi/6)}{n^2}$$

39 Complete the proof of the Ratio Test (11.31) by showing that if $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = \infty$, then $\sum a_n$ is divergent.

40 Complete the proof of the Root Test (11.32).

41 (a) Prove that if $\sum a_n$ is absolutely convergent, then $\sum a_n^2$ is convergent.

(b) Show that the conclusion of part (a) is not necessarily true if $\sum a_n$ is convergent.

(c) Show that if $\sum a_n^2$ is convergent, it does not necessarily follow that $\sum a_n$ is convergent.

42 Prove that if $\sum a_n$ is absolutely convergent and $a_n \neq 0$ for all n , then $\sum 1/|a_n|$ is divergent.

11.6 Power Series

In the previous sections we concentrated on infinite series with constant terms. Of major importance in applications are series whose terms contain variables, as in the next definition.

Definition (11.33)

Let x be a variable. A **power series in x** is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots$$

where each a_i is a real number.

If a number is substituted for x in Definition (11.33) we obtain a series of constant terms which may be tested for convergence or divergence. To simplify the general term of the power series it is assumed that $x^0 = 1$ even if $x = 0$. The main objective of this section is to determine all values of x for which a power series converges. Evidently, every power series in x converges if $x = 0$. To find other numbers that produce a convergent series we often employ the Ratio Test, as illustrated in the following examples.

Example 1 Find all values of x for which the following power series is absolutely convergent:

$$1 + \frac{1}{5}x + \frac{2}{5^2}x^2 + \cdots + \frac{n}{5^n}x^n + \cdots$$

Solution If we let $u_n = (n/5^n)x^n = nx^n/5^n$, then

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)x^{n+1}}{5^{n+1}} \cdot \frac{5^n}{nx^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)x}{5n} \right| = \lim_{n \rightarrow \infty} \left(\frac{n+1}{5n} \right) |x| = \frac{1}{5} |x|.\end{aligned}$$

By the Ratio Test, the series is absolutely convergent if the following equivalent inequalities are true:

$$\frac{1}{5}|x| < 1, \quad |x| < 5, \quad -5 < x < 5.$$

The series diverges if $\frac{1}{5}|x| > 1$, that is, if $x > 5$ or $x < -5$. If $\frac{1}{5}|x| = 1$, the Ratio Test gives no information and hence the numbers 5 and -5 require special consideration. Substituting 5 for x in the power series we obtain

$$1 + 1 + 2 + 3 + \cdots + n + \cdots$$

which is divergent. If we let $x = -5$ we obtain

$$1 - 1 + 2 - 3 + \cdots + n(-1)^n + \cdots$$

which is also divergent. Consequently, the power series is absolutely convergent for every x in the open interval $(-5, 5)$ and diverges elsewhere. ■

Example 2 Find all values of x for which the following power series is absolutely convergent:

$$1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \cdots + \frac{1}{n!}x^n + \cdots$$

Solution We shall employ the same technique used in Example 1. If we let

$$u_n = \frac{1}{n!}x^n = \frac{x^n}{n!}$$

$$\begin{aligned}\text{then } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{x}{n+1} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} |x| = 0.\end{aligned}$$

Since the limit 0 is less than 1 for every value of x , it follows from the Ratio Test that the power series is absolutely convergent for all real numbers. ■

Example 3 Find all values of x for which $\sum n!x^n$ is convergent.

Solution Let $u_n = n!x^n$. If $x \neq 0$, then

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)!x^{n+1}}{n!x^n} \right| \\ &= \lim_{n \rightarrow \infty} |(n+1)x| = \lim_{n \rightarrow \infty} (n+1)|x| = \infty\end{aligned}$$

and, by the Ratio Test, the series diverges. Hence the power series is convergent only if $x = 0$. ■

It will be shown in Theorem (11.35) that the solutions of the preceding examples are typical, in the sense that if a power series converges for nonzero values of x , then either it is absolutely convergent for all real numbers, or it is absolutely convergent throughout some open interval $(-r, r)$ and diverges outside of the closed interval $[-r, r]$. The proof of this fact depends on the next theorem.

Theorem (11.34)

- (i) If a power series $\sum a_n x^n$ converges for a nonzero number c , then it is absolutely convergent whenever $|x| < |c|$.
- (ii) If a power series $\sum a_n x^n$ diverges for a nonzero number d , then it diverges whenever $|x| > |d|$.

Proof If $\sum a_n c^n$ converges and $c \neq 0$, then by Theorem (11.14), $\lim_{n \rightarrow \infty} a_n c^n = 0$. Employing Definition (11.3) with $\varepsilon = 1$, there is a positive integer N such that

$$|a_n c^n| < 1 \quad \text{whenever } n \geq N$$

and, therefore,

$$|a_n x^n| = \left| \frac{a_n c^n x^n}{c^n} \right| = |a_n c^n| \left| \frac{x}{c} \right|^n < \left| \frac{x}{c} \right|^n$$

provided $n \geq N$. If $|x| < |c|$, then $|x/c| < 1$ and $\sum |x/c|^n$ is a convergent geometric series. Hence, by the Comparison Test (11.24), the series obtained by deleting the first N terms of $\sum |a_n x^n|$ is convergent. It follows that the series $\sum |a_n x^n|$ is also convergent, which proves (i).

To prove (ii), suppose the series diverges for $x = d \neq 0$. If the series converges for some c_1 , where $|c_1| > |d|$, then by (i) it converges whenever $|x| < |c_1|$. In particular it converges for $x = d$, contrary to our supposition. Hence the series diverges whenever $|x| > |d|$. □

Theorem (11.35)

If $\sum a_n x^n$ is a power series, then precisely one of the following is true.

- (i) The series converges only if $x = 0$.
- (ii) The series is absolutely convergent for all x .
- (iii) There is a positive number r such that the series is absolutely convergent if $|x| < r$ and divergent if $|x| > r$.

Proof If neither (i) nor (ii) is true, then there exist nonzero numbers c and d such that the series converges if $x = c$ and diverges if $x = d$. Let S denote the set of all real numbers for which the series is absolutely convergent. By Theorem (11.34), the series diverges if $|x| > |d|$ and hence every number in S is less than $|d|$. By the Completeness Property (11.9), S has a least upper bound r . It follows that the series is absolutely convergent if $|x| < r$ and diverges if $|x| > r$. $\square$

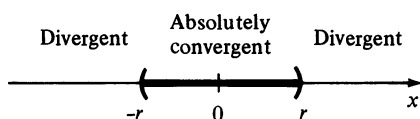


FIGURE 11.4 $\sum a_n x^n$, radius of convergence r .

If (iii) of Theorem (11.35) occurs, then the power series $\sum a_n x^n$ is absolutely convergent throughout the open interval $(-r, r)$ and diverges outside of the closed interval $[-r, r]$, as illustrated in Figure 11.4. The number r is called the **radius of convergence** of the series. Either convergence or divergence may occur at $-r$ or r , depending on the nature of the series.

The totality of numbers for which a power series converges is called its **interval of convergence**. If the radius of convergence r is positive, then the interval of convergence is one of the following:

$$(-r, r), \quad (-r, r], \quad [-r, r), \quad \text{or} \quad [-r, r].$$

If (i) or (ii) of Theorem (11.35) occurs, then the radius of convergence is denoted by 0 or ∞ , respectively. In Example 1 of this section, the interval of convergence is $(-5, 5)$ and the radius of convergence is 5. In Example 2 the interval of convergence is $(-\infty, \infty)$ and we write $r = \infty$. In Example 3, $r = 0$. The next example illustrates the case of a half-open interval of convergence.

Example 4 Find the interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} x^n$.

Solution Note that in this example the coefficient of x^0 is 0 and the summation begins with $n = 1$. We let $u_n = x^n/\sqrt{n}$ and consider

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{x^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\sqrt{n}}{\sqrt{n+1}} x \right| \\ &= \lim_{n \rightarrow \infty} \sqrt{\frac{n}{n+1}} |x| = (1)|x| = |x|. \end{aligned}$$

It follows from the Ratio Test that the power series is absolutely convergent if $|x| < 1$, that is, if $-1 < x < 1$. The series diverges if $|x| > 1$, that is, if $x > 1$

or $x < -1$. The numbers 1 and -1 must be checked by direct substitution in the power series. If we let $x = 1$ we obtain

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} (1)^n = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} + \cdots$$

which is a divergent p -series. If we substitute $x = -1$ the result is

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} (-1)^n = -1 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} + \cdots + \frac{(-1)^n}{\sqrt{n}} + \cdots$$

which converges by the Alternating Series Test. Hence the power series converges if $-1 \leq x < 1$, and the interval of convergence is $[-1, 1)$. ■

Definition (11.36)

Let c be a real number and x a variable. A **power series in $x - c$** is a series of the form

$$\sum_{n=0}^{\infty} a_n(x - c)^n = a_0 + a_1(x - c) + a_2(x - c)^2 + \cdots + a_n(x - c)^n + \cdots$$

where each a_i is a real number.

To simplify the n th term in (11.36), it is assumed that $(x - c)^0 = 1$ even if $x = c$. If we employ the same reasoning used to prove Theorem (11.35), and replace x by $x - c$, then it can be shown that precisely one of the following is true.

- (i) The series converges only if $x - c = 0$, that is, if $x = c$.
- (ii) The series is absolutely convergent for all x .
- (iii) There is a positive number r such that the series is absolutely convergent if $|x - c| < r$ and divergent if $|x - c| > r$.

If (iii) occurs, then the series $\sum a_n(x - c)^n$ is absolutely convergent if

$$-r < x - c < r, \quad \text{or} \quad c - r < x < c + r$$

that is, if x is the interval $(c - r, c + r)$ as illustrated in Figure 11.5. The endpoints of the interval must be checked separately. As before, the totality of numbers for which the series converges is called the **interval of convergence**, and r is called the **radius of convergence**.

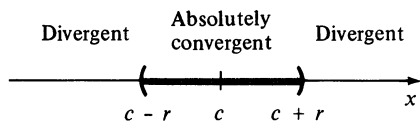


FIGURE 11.5 $\sum a_n(x - c)^n$, radius of convergence r .

Example 5 Find the interval of convergence of

$$1 - \frac{1}{2}(x - 3) + \frac{1}{3}(x - 3)^2 + \cdots + (-1)^n \frac{1}{n + 1}(x - 3)^n + \cdots$$

Solution If we let $u_n = (-1)^n(x - 3)^n/(n + 1)$, then

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(x - 3)^{n+1}}{n + 2} \cdot \frac{n + 1}{(x - 3)^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n + 1}{n + 2} (x - 3) \right| \\ &= \lim_{n \rightarrow \infty} \left(\frac{n + 1}{n + 2} \right) |x - 3| \\ &= (1)|x - 3| = |x - 3|.\end{aligned}$$

By the Ratio Test the series is absolutely convergent if $|x - 3| < 1$, that is, if

$$-1 < x - 3 < 1 \quad \text{or} \quad 2 < x < 4.$$

The series diverges if $x < 2$ or $x > 4$. The numbers 2 and 4 must be checked separately. If $x = 4$ the resulting series is

$$1 - \frac{1}{2} + \frac{1}{3} - \cdots + (-1)^n \frac{1}{n + 1} + \cdots$$

which converges by the Alternating Series Test. For $x = 2$ the series becomes

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n + 1} + \cdots$$

which is the divergent harmonic series. Hence the interval of convergence is $(2, 4]$. ■

11.6 Exercises

Find the interval of convergence of the power series in Exercises 1–26.

1 $\sum_{n=0}^{\infty} \frac{1}{n+4} x^n$

2 $\sum_{n=0}^{\infty} \frac{1}{n^2+4} x^n$

13 $\sum_{n=0}^{\infty} \frac{n!}{100^n} x^n$

14 $\sum_{n=0}^{\infty} \frac{(3n)!}{(2n)!} x^n$

3 $\sum_{n=0}^{\infty} \frac{n^2}{2^n} x^n$

4 $\sum_{n=1}^{\infty} \frac{(-3)^n}{n} x^{n+1}$

15 $\sum_{n=0}^{\infty} \frac{1}{(-4)^n} x^{2n+1}$

16 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\sqrt[3]{n} 3^n} x^n$

5 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\sqrt{n}} x^n$

6 $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)} x^n$

17 $\sum_{n=0}^{\infty} \frac{2^n}{(2n)!} x^{2n}$

18 $\sum_{n=0}^{\infty} \frac{10^n}{n!} x^n$

7 $\sum_{n=2}^{\infty} \frac{n}{n^2+1} x^n$

8 $\sum_{n=1}^{\infty} \frac{1}{4^n \sqrt{n}} x^n$

19 $\sum_{n=0}^{\infty} \frac{3^{2n}}{n+1} (x-2)^n$

20 $\sum_{n=1}^{\infty} \frac{1}{n 5^n} (x-5)^n$

9 $\sum_{n=2}^{\infty} \frac{\ln n}{n^3} x^n$

10 $\sum_{n=0}^{\infty} \frac{10^{n+1}}{3^{2n}} x^n$

21 $\sum_{n=0}^{\infty} \frac{n^2}{2^{3n}} (x+4)^n$

22 $\sum_{n=0}^{\infty} \frac{1}{2n+1} (x+3)^n$

11 $\sum_{n=0}^{\infty} \frac{n+1}{10^n} (x-4)^n$

12 $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} (x-2)^n$

23 $\sum_{n=1}^{\infty} \frac{\ln n}{e^n} (x-e)^n$

24 $\sum_{n=0}^{\infty} \frac{n}{3^{2n-1}} (x-1)^{2n}$

25 $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n 6^n} (2x-1)^n$

26 $\sum_{n=0}^{\infty} \frac{1}{\sqrt{3n+4}} (3x+4)^n$

Find the radius of convergence of the power series in Exercises 27–30.

$$27 \quad \sum_{n=1}^{\infty} (-1)^n \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{3 \cdot 6 \cdot 9 \cdots (3n)} x^n$$

$$28 \quad \sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{4 \cdot 7 \cdot 10 \cdots (3n+1)} x^n$$

$$29 \quad \sum_{n=1}^{\infty} \frac{n^n}{n!} x^n$$

$$30 \quad \sum_{n=0}^{\infty} \frac{(n+1)!}{10^n} (x-5)^n$$

Find the radius of convergence of the power series in Exercises 31 and 32, where c and d are positive integers.

$$31 \quad \sum_{n=0}^{\infty} \frac{(n+c)!}{n!(n+d)!} x^n$$

$$32 \quad \sum_{x=0}^{\infty} \frac{(cn)!}{(n!)^c} x^n$$

33 If $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = k$, where $k \neq 0$, prove that the radius of convergence of $\sum a_n x^n$ is $1/k$.

34 If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = k$, where $k \neq 0$, prove that the radius of convergence of $\sum a_n x^n$ is $1/k$.

35 If $\sum a_n x^n$ has radius of convergence r , prove that $\sum a_n x^{2n}$ has radius of convergence $\sqrt{r}$.

36 If $\sum a_n$ is absolutely convergent, prove that $\sum a_n x^n$ is absolutely convergent for all x in the interval $[-1, 1]$.

37 If the interval of convergence of $\sum a_n x^n$ is $(-r, r]$, prove that the series is conditionally convergent at r .

38 If $\sum a_n x^n$ is absolutely convergent at one endpoint of its interval of convergence, prove that it is also absolutely convergent at the other endpoint.

11.7 Power Series Representations of Functions

A power series $\sum a_n x^n$ may be used to define a function f whose domain is the interval of convergence of the series. Specifically, for each x in this interval we let $f(x)$ equal the sum of the series, that is,

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots$$

If a function f is defined in this way we say that $\sum a_n x^n$ is a **power series representation for $f(x)$** (or of $f(x)$). We also use the phrase *f is represented by the power series*.

A power series representation $f(x) = \sum a_n x^n$ enables us to find functional values in a new way. Specifically, if c is in the interval of convergence, then $f(c)$ can be found (or approximated) by finding (or approximating) the sum of the series

$$a_0 + a_1 c + a_2 c^2 + \cdots + a_n c^n + \cdots$$

Power series representations will also enable us to solve problems involving differentiation and integration by using techniques different from those considered earlier in the text.

Example 1 Find a function f that is represented by the power series

$$\sum_{n=0}^{\infty} (-1)^n x^n.$$

Solution If $|x| < 1$, then by Theorem (11.13), the geometric series

$$1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots$$

converges and has the sum $1/[1 - (-x)] = 1/(1 + x)$, and hence we may write

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots$$

This gives us a power series representation for $f(x) = 1/(1 + x)$, if $|x| < 1$. ■

Most of the examples and exercises in this section are concerned with geometric series, since the sum can be determined by means of a specific formula. In the next section we shall consider the following more difficult problem: Given a function f , find a power series representation for f .

A function f defined by a power series has properties similar to those of a polynomial. In particular, it can be shown that f has a derivative whose power series representation may be found by differentiating each term of the given series. Similarly, definite integrals of f may be obtained by integrating each term of the series. These facts are consequences of the next theorem, which is stated without proof.

Theorem (11.37)

Suppose a power series $\sum a_n x^n$ has a nonzero radius of convergence r and let the function f be defined by

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots$$

for every x in the interval of convergence. If $-r < x < r$, then

$$\begin{aligned} \text{(i)} \quad f'(x) &= \sum_{n=0}^{\infty} D_x(a_n x^n) = \sum_{n=1}^{\infty} n a_n x^{n-1} \\ &= a_1 + 2a_2 x + 3a_3 x^2 + \cdots + n a_n x^{n-1} + \cdots \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int_0^x f(t) dt &= \sum_{n=0}^{\infty} \int_0^x (a_n t^n) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \\ &= a_0 x + \frac{1}{2} a_1 x^2 + \frac{1}{3} a_2 x^3 + \cdots + \frac{1}{n+1} a_n x^{n+1} + \cdots \end{aligned}$$

It can be shown that the series in (i) and (ii) of Theorem (11.37) have the same radius of convergence as $\sum a_n x^n$. As a corollary of (i), *a function that is represented by a power series in an interval $(-r, r)$ is continuous throughout $(-r, r)$.* (See Theorem (3.8).) Similar results are true for functions represented by power series of the form $\sum a_n (x - c)^n$.

Example 2 Find a power series representation for $1/(1 + x)^2$.

Solution It was shown in Example 1 that

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots$$

If we differentiate each term of this series, then by (i) of Theorem (11.37),

$$-\frac{1}{(1+x)^2} = -1 + 2x - 3x^2 - \cdots + (-1)^n nx^{n-1} + \cdots$$

Consequently, if $|x| < 1$, then

$$\frac{1}{(1+x)^2} = 1 - 2x + 3x^2 + \cdots + (-1)^{n+1} nx^{n-1} + \cdots \quad \blacksquare$$

Example 3 Find a power series representation for $\ln(1+x)$ if $|x| < 1$.

Solution If $|x| < 1$, then

$$\begin{aligned} \ln(1+x) &= \int_0^x \frac{1}{1+t} dt \\ &= \int_0^x [1 - t + t^2 - \cdots + (-1)^n t^n + \cdots] dt \end{aligned}$$

where the last equality follows from Example 1. By (ii) of Theorem (11.37), we may integrate each term of the series as follows:

$$\begin{aligned} \ln(1+x) &= \int_0^x 1 dt - \int_0^x t dt + \int_0^x t^2 dt - \cdots + (-1)^n \int_0^x t^n dt + \cdots \\ &= \left[t - \frac{t^2}{2} + \frac{t^3}{3} - \cdots + (-1)^n \frac{t^{n+1}}{n+1} \right]_0^x + \cdots \end{aligned}$$

Hence

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^n \frac{x^{n+1}}{n+1} + \cdots$$

if $|x| < 1$. ■

Example 4 Calculate $\ln(1.1)$ to five decimal places.

Solution In Example 3 we found a series representation for $\ln(1+x)$ if $|x| < 1$. Substituting 0.1 for x in that series gives us

$$\begin{aligned} \ln(1.1) &= 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} + \frac{(0.1)^5}{5} - \cdots \\ &= 0.1 - 0.005 + 0.000333 - 0.000025 + 0.000002 - \cdots \end{aligned}$$

If we sum the first four terms on the right and round off to five decimal places, then

$$\ln(1.1) \approx 0.09531.$$

By Theorem (11.27), the error is less than the absolute value 0.000002 of the fifth term and, therefore, the number 0.09531 is accurate to five decimal places. The reader should compare this solution to that of Example 2 in Section 10.5. ■

Example 5 Find a power series representation for $\arctan x$.

Solution We first observe that

$$\arctan x = \int_0^x \frac{1}{1+t^2} dt.$$

Next we note that if $|t| < 1$, then by Theorem (11.13), with $a = 1$ and $r = -t^2$,

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - \cdots + (-1)^n t^{2n} + \cdots$$

By (ii) of Theorem (11.37) we may integrate each term of the series to obtain

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots$$

provided $|x| < 1$. It can be proved, by more advanced methods, that this series representation is also valid if $|x| = 1$. ■

Example 6 Prove that e^x has the power series representation

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$$

Solution The indicated power series was considered in Example 2 of the preceding section, where it was shown to be absolutely convergent for every real number x . If we let f denote the function represented by the series, then

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Applying (i) of Theorem (11.37),

$$\begin{aligned} f'(x) &= \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} \\ &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \end{aligned}$$

that is,

$$f'(x) = f(x) \quad \text{for every } x.$$

If, in Theorem (7.25), we let $q = f$, $t = x$, and $c = 1$, we obtain

$$f(x) = f(0)e^x.$$

However,

$$f(0) = 1 + 0 + \frac{0^2}{2!} + \cdots + \frac{0^n}{n!} + \cdots = 1$$

and hence

$$f(x) = e^x$$

which is what we wished to prove. ■

Note that Example 6 allows us to express the number e as the sum of a convergent positive term series, namely,

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \cdots$$

Example 7 Approximate $\int_0^{0.1} e^{-x^2} dx$.

Solution Letting $x = -t^2$ in the series of Example 6 gives us

$$e^{-t^2} = 1 - t^2 + \frac{t^4}{2!} - \cdots + \frac{(-1)^n t^{2n}}{n!} + \cdots$$

which is true for all t . Applying (ii) of Theorem (11.37),

$$\begin{aligned} \int_0^{0.1} e^{-x^2} dx &= \int_0^{0.1} e^{-t^2} dt \\ &= t \Big|_0^{0.1} - \frac{t^3}{3} \Big|_0^{0.1} + \frac{t^5}{10} \Big|_0^{0.1} - \cdots \\ &= 0.1 - \frac{(0.1)^3}{3} + \frac{(0.1)^5}{10} - \cdots \end{aligned}$$

If we use the first two terms to approximate the sum of this convergent alternating series, then by Theorem (11.27), the error is less than the third term $(0.1)^5/10 = 0.000001$. Hence

$$\int_0^{0.1} e^{-x^2} dx \approx 0.1 - \frac{0.001}{3} \approx 0.99667$$

which is accurate to five decimal places. ■

Thus far, the techniques used to obtain power series representations of functions were *indirect*, in the sense that we started with known series and then differentiated or integrated, and perhaps used other knowledge about functional behavior, to obtain series representations. In the next section we shall discuss a *direct* method which can be used to find power series representations for a large variety of functions.

11.7 Exercises

In Exercises 1–10, find a power series representation for $f(x)$ and specify the interval of convergence.

1 $f(x) = 1/(1 - x)$

2 $f(x) = 1/(1 + x^2)$

3 $f(x) = 1/(1 - x)^2$

4 $f(x) = 1/(1 - 4x)$

5 $f(x) = x^2/(1 - x^2)$

7 $f(x) = x/(2 - 3x)$

9 $f(x) = (x^2 + 1)/(x - 1)$

6 $f(x) = x/(1 - x^4)$

8 $f(x) = x^3/(4 - x^3)$

10 $f(x) = 3/(2x + 5)$

11 Prove that

$$\ln(1-x) = \sum_{n=1}^{\infty} \frac{-x^n}{n} \quad \text{if } |x| < 1,$$

and use this fact to approximate $\ln(1.2)$ to three decimal places.

12 Use Example 5 to prove that the sum of the series $\sum_{n=0}^{\infty} (-\frac{1}{3})^n / \sqrt{3}(2n+1)$ is $\pi/6$.

In Exercises 13–18 use the result obtained in Example 6 to find a power series representation for $f(x)$.

13 $f(x) = e^{-x}$

14 $f(x) = e^{2x}$

15 $f(x) = \cosh x$

16 $f(x) = \sinh x$

17 $f(x) = xe^{3x}$

18 $f(x) = x^2e^{x^2}$

Use infinite series to approximate each of the integrals in Exercises 19–24 to four decimal places.

19 $\int_0^{1/3} \frac{1}{1+x^6} dx$

20 $\int_0^{1/2} \arctan x^2 dx$

21 $\int_{0.1}^{0.2} \frac{\arctan x}{x} dx$

22 $\int_0^{0.2} \frac{x^3}{1+x^5} dx$

23 $\int_0^1 e^{-x^2/10} dx$

24 $\int_0^{0.5} e^{-x^3} dx$

25 Use the power series representation for $(1-x^2)^{-1}$ to find a power series representation for $2x(1-x^2)^{-2}$.

26 Use the method of Example 3 to find a power series representation for $\ln(3+2x)$.

In Exercises 27–30 find a power series representation for $f(x)$. If the integrand is denoted by $g(t)$, assume that the value of $g(0)$ is $\lim_{t \rightarrow 0} g(t)$. (Why is a condition of this type necessary?)

27 $f(x) = \int_0^x \frac{e^{-t^2} - 1}{t} dt$

28 $f(x) = \int_0^x \frac{e^t - 1}{t} dt$

29 $f(x) = \int_0^x \frac{\ln(1-t)}{t} dt$

30 $f(x) = \int_0^x \ln(1+t^2) dt$

11.8 Taylor and Maclaurin Series

Suppose a function f is represented by a power series in $x - c$, such that

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} a_n(x-c)^n \\ &= a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + a_4(x-c)^4 + \cdots \end{aligned}$$

where the domain of f is an open interval containing c . As in the preceding section, power series representations may be found for $f'(x)$, $f''(x)$, $\dots$, by differentiating the terms of the series. Thus

$$\begin{aligned} f'(x) &= \sum_{n=1}^{\infty} n a_n(x-c)^{n-1} \\ &= a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + 4a_4(x-c)^3 + \cdots \end{aligned}$$

$$\begin{aligned} f''(x) &= \sum_{n=2}^{\infty} n(n-1)a_n(x-c)^{n-2} \\ &= 2a_2 + (3 \cdot 2)a_3(x-c) + (4 \cdot 3)a_4(x-c)^2 + \cdots \end{aligned}$$

$$\begin{aligned} f'''(x) &= \sum_{n=3}^{\infty} n(n-1)(n-2)a_n(x-c)^{n-3} \\ &= (3 \cdot 2)a_3 + (4 \cdot 3 \cdot 2)a_4(x-c) + \cdots \end{aligned}$$

and, for every positive integer k ,

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)a_n(x-c)^{n-k}.$$

Moreover, each series obtained by differentiation has the same radius of convergence as the original series. Substituting c for x in each of these series representations, we obtain

$$f(c) = a_0, \quad f'(c) = a_1, \quad f''(c) = 2a_2, \quad f'''(c) = (3 \cdot 2)a_3$$

and, for every positive integer n ,

$$f^{(n)}(c) = n!a_n, \quad \text{or} \quad a_n = \frac{f^{(n)}(c)}{n!}.$$

We have proved the following result.

Theorem (11.38)

If f is a function and

$$f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$$

for all x in an open interval containing c , then

$$\begin{aligned} f(x) &= f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots \\ &\quad + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots \end{aligned}$$

The series that appears in the conclusion of Theorem (11.38) is called the **Taylor series for $f(x)$ at c** . The special case $c = 0$, stated in the following corollary, is extremely important.

Corollary (11.39)

If f is a function and $f(x) = \sum a_n x^n$ for all x in an open interval $(-r, r)$, then

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

The series in this corollary is called the **Maclaurin series for $f(x)$** . Each example in the preceding section involves a Maclaurin series.

Theorem (11.38) states that if a function f is represented by a power series in $x - c$, then the power series *must* be the Taylor series. However, the theorem does not state conditions which guarantee that a power series

representation actually exists. We shall now obtain such conditions. Let us begin by noting that the $(n + 1)$ st partial sum of the Taylor series stated in (11.38) is the n th-degree Taylor Polynomial $P_n(x)$ of f at c (see Definition (10.9)). Moreover, by Theorem (10.10),

$$P_n(x) = f(x) - R_n(x)$$

where
$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!} (x - c)^{n+1}$$

for some number z between c and x . In the next theorem we use $R_n(x)$ to specify sufficient conditions for the existence of a power series representation of $f(x)$.

Theorem (11.40)

If a function f has derivatives of all orders throughout an interval containing c , and if

$$\lim_{n \rightarrow \infty} R_n(x) = 0$$

for every x in that interval, then $f(x)$ is represented by the Taylor series for $f(x)$ at c .

Proof The polynomial $P_n(x)$ is a general term for the sequence of partial sums of the Taylor series for $f(x)$ at c . Moreover, since $P_n(x) = f(x) - R_n(x)$,

$$\lim_{n \rightarrow \infty} P_n(x) = f(x) - \lim_{n \rightarrow \infty} R_n(x) = f(x).$$

Hence the sequence of partial sums converges to $f(x)$. This proves the theorem. $\square$

In Example 2 of Section 11.6 we proved that the power series $\sum x^n/n!$ is absolutely convergent for all real numbers. Using Theorem (11.14) gives us the following result.

Theorem (11.41)

For every real number x ,

$$\lim_{n \rightarrow \infty} \left| \frac{x^n}{n!} \right| = 0.$$

We shall use Theorem (11.41) in the solution of the following example.

Example 1 Find the Maclaurin series for $\sin x$ and prove that it represents $\sin x$ for every real number x .

Solution Let us arrange our work as follows:

$$\begin{aligned} f(x) &= \sin x & f(0) &= 0 \\ f'(x) &= \cos x & f'(0) &= 1 \\ f''(x) &= -\sin x & f''(0) &= 0 \\ f'''(x) &= -\cos x & f'''(0) &= -1 \end{aligned}$$

Successive derivatives follow this same pattern. Substitution in Corollary (11.39) yields the Maclaurin series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots$$

At this stage all we know is that if $\sin x$ is represented by a power series in x , then it is given by the preceding series. To prove that the representation is actually true for every number x , let us employ Theorem (11.40). If n is a positive integer, then either

$$|f^{(n+1)}(x)| = |\cos x| \quad \text{or} \quad |f^{(n+1)}(x)| = |\sin x|.$$

Hence $|f^{(n+1)}(z)| \leq 1$ for every number z and, using the formula for $R_n(x)$ with $c = 0$,

$$|R_n(x)| = \frac{|f^{(n+1)}(z)|}{(n+1)!} |x|^{n+1} \leq \frac{|x|^{n+1}}{(n+1)!}.$$

It follows from Theorem (11.41) and the Sandwich Theorem (11.6) that $\lim_{n \rightarrow \infty} |R_n(x)| = 0$. Consequently, $\lim_{n \rightarrow \infty} R_n(x) = 0$, and the Maclaurin series representation for $\sin x$ is true for every x . ■

Example 2 Find the Maclaurin series for $\cos x$.

Solution We could proceed directly as in Example 1; however, let us obtain the desired series by differentiating the series for $\sin x$ obtained in Example 1. This gives us

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots$$

This series can also be obtained by integrating the terms of the series for $\sin x$. ■

The Maclaurin series for e^x was obtained in Example 6 of the preceding section by using an indirect technique. We shall next give a direct derivation of this important formula.

Example 3 Find a Maclaurin series which represents e^x for every real number x .

Solution If $f(x) = e^x$, then $f^{(n)}(x) = e^x$ for every positive integer n . Hence $f^{(n)}(0) = 1$ and substitution in Corollary (11.39) gives us

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$$

As in the solution of Example 1, we now employ Theorem (11.40) to prove that this power series representation of e^x is true for every real number x . Using the formula for $R_n(x)$ with $c = 0$,

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!} x^{n+1} = \frac{e^z}{(n+1)!} x^{n+1}$$

where z is a number between 0 and x . If $0 < x$, then $e^z < e^x$ since the natural exponential function is increasing, and hence, for every positive integer n ,

$$0 < R_n(x) < \frac{e^x}{(n+1)!} x^{n+1}.$$

Using Theorem (11.41),

$$\lim_{n \rightarrow \infty} \frac{e^x}{(n+1)!} x^{n+1} = e^x \lim_{n \rightarrow \infty} \frac{x^{n+1}}{(n+1)!} = 0$$

and, by the Sandwich Theorem (11.6),

$$\lim_{n \rightarrow \infty} R_n(x) = 0.$$

If $x < 0$, then also $z < 0$ and hence $e^z < e^0 = 1$. Consequently,

$$0 < |R_n(x)| < \left| \frac{x^{n+1}}{(n+1)!} \right|$$

and we again see that $R_n(x)$ has the limit 0 as n increases without bound. It follows from Theorem (11.40) that the power series representation for e^x is valid for all nonzero x . Finally, note that if $x = 0$, then the series reduces to $e^0 = 1$. ■

Example 4 Find the Taylor series for $\sin x$ in powers of $x - \pi/6$.

Solution The derivatives of $f(x) = \sin x$ are listed in Example 1. If we evaluate them at $c = \pi/6$, we obtain

$$f\left(\frac{\pi}{6}\right) = \frac{1}{2}, \quad f'\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad f''\left(\frac{\pi}{6}\right) = -\frac{1}{2}, \quad f'''\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2},$$

and this pattern of four numbers repeats itself indefinitely. Substitution in Theorem (11.38) gives us

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6}\right) - \frac{1}{2(2!)} \left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{2(3!)} \left(x - \frac{\pi}{6}\right)^3 + \cdots$$

The general term u_n of this series is given by

$$u_n = \begin{cases} (-1)^{n/2} \frac{1}{2n!} \left(x - \frac{\pi}{6}\right)^n & \text{if } n = 0, 2, 4, 6, \dots \\ (-1)^{(n-1)/2} \frac{\sqrt{3}}{2n!} \left(x - \frac{\pi}{6}\right)^n & \text{if } n = 1, 3, 5, 7, \dots \end{cases}$$

The proof that the series represents $\sin x$ for all x is similar to that given in Example 1 and is, therefore, omitted. ■

The next example brings out the fact that a function f may have derivatives of all orders at some number c , but may not have a Taylor series representation at that number. This shows that an additional condition, such as $\lim_{n \rightarrow \infty} R_n(x) = 0$, is required to guarantee the existence of a Taylor series.

Example 5 Let f be the function defined by

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Show that $f(x)$ cannot be represented by a Maclaurin series.

Solution Using Theorem (3.7), the derivative of f at 0 is

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x} = \lim_{x \rightarrow 0} \frac{(1/x)}{e^{1/x^2}}.$$

The last expression has the indeterminate form ∞/∞ . Applying L'Hôpital's Rule we see that

$$f'(0) = \lim_{x \rightarrow 0} \frac{(-1/x^2)}{(-2/x^3)e^{1/x^2}} = \lim_{x \rightarrow 0} \frac{x}{2e^{1/x^2}} = 0.$$

It can be proved that $f''(0) = 0$, $f'''(0) = 0$ and, in general, $f^{(n)}(0) = 0$ for every positive integer n . According to Corollary (11.39), if $f(x)$ has a Maclaurin series representation, then it is given by

$$f(x) = 0 + 0x + \frac{0}{2!}x^2 + \dots + \frac{0}{n!}x^n + \dots$$

which implies that $f(x) = 0$ throughout an interval containing 0. However, this contradicts the definition of f . Consequently, $f(x)$ does not have a Maclaurin series representation. ■

As a by-product of Example 5, it follows from Theorem (11.40) that for the given function f , $\lim_{x \rightarrow \infty} R_n(x) \neq 0$ at $c = 0$.

In previous examples we established some of the Maclaurin series given in the following list. Verifications of the remaining series are left as exercises.

(11.42)	<i>Important Maclaurin Series</i>	<i>Interval of Convergence</i>
(a)	$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots$	$(-\infty, \infty)$
(b)	$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots$	$(-\infty, \infty)$
(c)	$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$	$(-\infty, \infty)$
(d)	$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^n \frac{x^{n+1}}{n+1} + \cdots$	$(-1, 1]$
(e)	$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots$	$[-1, 1]$
(f)	$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + \cdots$	$(-\infty, \infty)$
(g)	$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \cdots$	$(-\infty, \infty)$

The series in (11.42) can be used to obtain power series representations for other functions. To illustrate, since (c) is true for every x , a power series representation for e^{-x^2} can be found by substituting $-x^2$ for x . This gives us

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \cdots + (-1)^n \frac{x^{2n}}{n!} + \cdots$$

Similarly, replacing x by $-x$ in (c) leads to

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \cdots + (-1)^n \frac{x^n}{n!} + \cdots$$

Since $\cosh x = \frac{1}{2}(e^x + e^{-x})$, a power series for $\cosh x$ can be found by adding corresponding terms of the series for e^x and e^{-x} , and then multiplying by $\frac{1}{2}$. This produces

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \cdots$$

It is interesting to compare this with series (b) for $\cos x$.

Taylor and Maclaurin series may be used to approximate values of functions in a manner similar to those used in the preceding section and in Section 10.5. For example, to find $\sin(0.1)$ we could use (a) of (11.42) to write

$$\sin(0.1) = 0.1 - \frac{0.001}{6} + \frac{0.00001}{120} - \cdots$$

By Theorem (11.27) the error involved in using the sum of the first two terms as an approximation is less than $0.00001/120 \approx 0.00000008$. More generally, we have the following polynomial approximation formula:

$$\sin x \approx x - \frac{x^3}{6}$$

where the error is less than $|x^5|/5!$. (Why?)

The next example illustrates the use of infinite series in approximating values of definite integrals.

Example 6 Approximate $\int_0^1 \sin x^2 dx$ to four decimal places.

Solution From (a) of (11.42),

$$\sin x^2 = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \frac{x^{14}}{7!} + \cdots$$

Integrating each term of the series, we obtain

$$\int_0^1 \sin x^2 dx = \frac{1}{3} - \frac{1}{42} + \frac{1}{1320} - \frac{1}{75600} + \cdots$$

Summing the first three terms,

$$\int_0^1 \sin x^2 dx \approx 0.3103$$

where the error is less than $1/75600 \approx 0.000013$. ■

Note that in the preceding example we achieved accuracy to four decimal places by summing only *three* terms of the integrated series for $\sin x^2$. To obtain this degree of accuracy by means of the Trapezoidal Rule or Simpson's Rule, it would be necessary to use an extremely fine partition of the interval $[0, 1]$. This brings out an important point. For numerical applications, in addition to analyzing a given problem we should also strive to find the most efficient method for computing the answer.

To obtain a Taylor or Maclaurin series representation for a function f by means of (11.38) or (11.39), respectively, it is necessary to find a general formula for $f^{(n)}(x)$ and, in addition, investigate $\lim_{n \rightarrow \infty} R_n(x)$. Because of this, our examples were restricted to expressions such as $\sin x$, $\cos x$, and e^x . The method cannot be used if, for example, $f(x)$ equals $\tan x$ or $\sin^{-1} x$, since $f^{(n)}(x)$ becomes very complicated as n increases. Most of the exercises that follow are based on functions whose n th derivatives can be determined easily, or on series representations that we have already established. In more complicated cases we shall restrict our attention to only the first few terms of a Taylor or Maclaurin series representation.

11.8 Exercises

In Exercises 1–4 use Corollary (11.39) to find the Maclaurin series for $f(x)$ and prove that the series is valid for all x by means of Theorem (11.40).

- 1 $f(x) = \cos x$ 2 $f(x) = e^{-x}$
 3 $f(x) = e^{2x}$ 4 $f(x) = \cosh x$

In Exercises 5–12 use (11.42) to find a Maclaurin series for $f(x)$ and state the radius of convergence.

- 5 $f(x) = x^2 e^x$
 6 $f(x) = x e^{-2x}$
 7 $f(x) = \sinh x$
 8 $f(x) = x^2 \sin x$
 9 $f(x) = x \sin 3x$
 10 $f(x) = \cos x^2$
 11 $f(x) = \cos^2 x$ (Hint: Use $\cos^2 x = (1 + \cos 2x)/2$.)
 12 $f(x) = \sin^2 x$

In Exercises 13–18 find the Taylor Series for $f(x)$ at the indicated number c . (Do not verify that $\lim_{n \rightarrow \infty} R_n(x) = 0$.)

- 13 $f(x) = \sin x$; $c = \pi/4$
 14 $f(x) = \cos x$; $c = \pi/3$
 15 $f(x) = 1/x$; $c = 2$
 16 $f(x) = e^x$; $c = -3$
 17 $f(x) = 10^x$; $c = 0$
 18 $f(x) = \ln(1+x)$; $c = 0$
 19 Find a series representation for e^{2x} in powers of $x + 1$.
 20 Find a series representation for $\ln x$ in powers of $x - 1$.

In Exercises 21–26, find the first four terms of the Taylor series for $f(x)$ at the indicated value of c .

- 21 $f(x) = \sec x$; $c = \pi/3$ 22 $f(x) = \tan x$; $c = \pi/4$
 23 $f(x) = \sin^{-1} x$; $c = 1/2$ 24 $f(x) = \csc x$; $c = 2\pi/3$
 25 $f(x) = x e^x$; $c = -1$ 26 $f(x) = \operatorname{sech} x$; $c = 0$

In Exercises 27–40, use an infinite series to approximate the given number to four decimal places.

- 27 $\sqrt{e}$ 28 e^{-1}
 29 $\cos 3^\circ$ 30 $\sin 1^\circ$
 31 $\tan^{-1} 0.1$ 32 $\ln 1.5$
 33 $\sinh 0.5$ 34 $\cosh 0.1$
 35 $\int_0^1 e^{-x^2} dx$ 36 $\int_0^{1/2} x \cos x^3 dx$
 37 $\int_0^{0.5} \cos x^2 dx$ 38 $\int_0^{0.1} \tan^{-1} x^2 dx$
 39 $\int_0^{0.25} \sqrt{x} \ln(1+x) dx$ 40 $\int_0^{0.2} \cos \sqrt{x} dx$

Approximate the integrals in Exercises 41–46 to four decimal places. Assume that if the integrand is $f(x)$, then $f(0) = \lim_{x \rightarrow 0} f(x)$.

- 41 $\int_0^1 \frac{1 - \cos x}{x^2} dx$ 42 $\int_0^1 \frac{\sin x}{x} dx$
 43 $\int_0^{1/2} \frac{\ln(1+x)}{x} dx$ 44 $\int_0^{0.3} \frac{\tan^{-1} x}{x} dx$
 45 $\int_{-1}^0 \frac{e^{3x} - 1}{10x} dx$ 46 $\int_0^1 \frac{1 - e^{-x}}{x} dx$

- 47 Use (d) of (11.42) to find the Maclaurin series for $\ln [(1+x)/(1-x)]$.
 48 Use the first five terms of the series in Exercise 47 to calculate $\ln 2$, and compare your answer to the value obtained using a table or calculator.
 49 Use (e) of (11.42), with $x = 1$, to represent π as the sum of an infinite series. What accuracy is obtained by using the first five terms of the series to approximate π ? Approximately how many terms of the series are required to obtain four decimal-place accuracy for π ?
 50 Use the identity $\tan^{-1}(1/2) + \tan^{-1}(1/3) = \pi/4$ (see Exercise 30 of Section 8.3) to express π as the sum of two infinite series. Use the first five terms of each series to approximate π and compare the result with that obtained in Exercise 49.

11.9 The Binomial Series

The Binomial Theorem (3.12) states that if k is a positive integer, then for all numbers a and b ,

$$(a + b)^k = a^k + ka^{k-1}b + \frac{k(k-1)}{2!}a^{k-2}b^2 + \cdots + \frac{k(k-1)\cdots(k-n+1)}{n!}a^{k-n}b^n + \cdots + b^k.$$

If we let $a = 1$ and $b = x$, then

$$(1 + x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \cdots + \frac{k(k-1)\cdots(k-n+1)}{n!}x^n + \cdots + x^k.$$

If k is not a positive integer (or 0), it is useful to study the power series $\sum a_n x^n$, where $a_0 = 1$ and $a_n = k(k-1)\cdots(k-n+1)/n!$ for $n \geq 1$. This series has the form

$$1 + kx + \frac{k(k-1)}{2!}x^2 + \cdots + \frac{k(k-1)\cdots(k-n+1)}{n!}x^n + \cdots$$

and is called the **Binomial Series**. If k is a nonnegative integer, the series reduces to the finite sum given in the Binomial Theorem. Otherwise the series does not terminate. It is left as an exercise to show that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}x^{n+1}}{a_n x^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{k-n}{n+1} \right| |x| = |x|.$$

Hence, by the Ratio Test, the series is absolutely convergent if $|x| < 1$ and is divergent if $|x| > 1$. Thus the Binomial Series represents a function f , where

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \quad \text{if } |x| < 1.$$

It has already been noted that if k is a nonnegative integer, then $f(x) = (1+x)^k$. We shall now prove that the same is true for *every* real number k . Differentiating the Binomial Series gives us

$$f'(x) = k + k(k-1)x + \cdots + \frac{nk(k-1)\cdots(k-n+1)}{n!}x^{n-1} + \cdots$$

and, therefore,

$$xf'(x) = kx + k(k-1)x^2 + \cdots + \frac{nk(k-1)\cdots(k-n+1)}{n!}x^n + \cdots$$

If we add corresponding terms of the last two power series, then the coefficient of x^n is

$$\frac{(n+1)k(k-1)\cdots(k-n)}{(n+1)!} + \frac{nk(k-1)\cdots(k-n+1)}{n!}$$

which simplifies to

$$[(k-n) + n] \frac{k(k-1)\cdots(k-n+1)}{n!} = ka_n.$$

Consequently,

$$f'(x) + xf'(x) = \sum_{n=0}^{\infty} ka_n x^n = kf(x)$$

and, therefore,

$$f'(x)(1+x) - kf(x) = 0.$$

If we define the function g by $g(x) = f(x)/(1+x)^k$, then

$$\begin{aligned} g'(x) &= \frac{(1+x)^k f'(x) - f(x)k(1+x)^{k-1}}{(1+x)^{2k}} \\ &= \frac{(1+x)f'(x) - kf(x)}{(1+x)^{k+1}} = 0. \end{aligned}$$

It follows from Theorem (4.33) that $g(x) = c$ for some constant c , that is,

$$\frac{f(x)}{(1+x)^k} = c.$$

Since $f(0) = 1$, we see that $c = 1$ and hence $f(x) = (1+x)^k$, which is what we wished to prove. The next statement summarizes this discussion.

Theorem (11.43)

If $|x| < 1$, then for every real number k ,

$$\begin{aligned} (1+x)^k &= 1 + kx + \frac{k(k-1)}{2!}x^2 + \cdots \\ &\quad + \frac{k(k-1)\cdots(k-n+1)}{n!}x^n + \cdots \end{aligned}$$

Example 1 Find a power series representation for $\sqrt[3]{1+x}$.

Solution Using Theorem (11.43) with $k = \frac{1}{3}$,

$$\begin{aligned} \sqrt[3]{1+x} &= 1 + \frac{1}{3}x + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!}x^2 + \frac{\frac{1}{3}(\frac{1}{3}-1)(\frac{1}{3}-2)}{3!}x^3 + \cdots \\ &\quad + \frac{\frac{1}{3}(\frac{1}{3}-1)\cdots(\frac{1}{3}-n+1)}{n!}x^n + \cdots \end{aligned}$$

which may be written

$$\begin{aligned}\sqrt[3]{1+x} &= 1 + \frac{1}{3}x - \frac{2}{3^2 \cdot 2!}x^2 + \frac{1 \cdot 2 \cdot 5}{3^3 \cdot 3!}x^3 + \cdots \\ &\quad + (-1)^{n+1} \frac{1 \cdot 2 \cdots (3n-4)}{3^n \cdot n!}x^n + \cdots\end{aligned}$$

where $|x| < 1$. The formula for the n th term of this series is valid provided $n \geq 2$. ■

Example 2 Find a power series representation for $\sqrt[3]{1+x^4}$.

Solution The desired series can be obtained by substituting x^4 for x in the series of Example 1. Hence, if $|x| < 1$, then

$$\begin{aligned}\sqrt[3]{1+x^4} &= 1 + \frac{1}{3}x^4 - \frac{2}{3^2 \cdot 2!}x^8 + \cdots \\ &\quad + (-1)^{n+1} \frac{1 \cdot 2 \cdots (3n-4)}{3^n \cdot n!}x^{4n} + \cdots\end{aligned}\quad \blacksquare$$

Example 3 Approximate $\int_0^{0.3} \sqrt[3]{1+x^4} dx$.

Solution Integrating the terms of the series obtained in Example 2 gives us

$$\int_0^{0.3} \sqrt[3]{1+x^4} dx = 0.3 + 0.000162 - 0.000000243 + \cdots$$

Consequently, the integral may be approximated by 0.300162, which is accurate to six decimal places since the error is less than 0.000000243. (Why?) ■

The Binomial Series can be used to obtain polynomial approximation formulas for $(1+x)^k$. To illustrate, if $|x| < 1$, then from Example 1

$$\sqrt[3]{1+x} \approx 1 + \frac{1}{3}x,$$

where the error involved in this approximation is less than the next term $x^2/9$.

11.9 Exercises

In Exercises 1–6 find a power series representation for $f(x)$ and state the radius of convergence.

$$\begin{array}{ll} 1 \quad (a) f(x) = \sqrt{1+x} & 2 \quad (a) f(x) = \frac{1}{\sqrt[3]{1+x}} \\ (b) f(x) = \sqrt{1-x^3} & (b) f(x) = \frac{1}{\sqrt[3]{1-x^2}} \end{array}$$

$$3 \quad f(x) = (1+x)^{-3} \qquad 4 \quad f(x) = x(1+2x)^{-2}$$

$$5 \quad f(x) = \sqrt[3]{8+x} \qquad 6 \quad f(x) = (4+x)^{3/2}$$

7 Obtain a power series representation for $\sin^{-1} x$ by using $\sin^{-1} x = \int_0^x 1/\sqrt{1-t^2} dt$. What is the radius of convergence?

8 Obtain a power series representation for $\sinh^{-1} x$ by using $\sinh^{-1} x = \int_0^x 1/\sqrt{1+t^2} dt$. What is the radius of convergence?

Approximate the integrals in Exercises 9 and 10 to three decimal places.

$$9 \quad \int_0^{1/2} \sqrt{1+x^3} dx \quad (\text{see Exercise 1})$$

$$10 \quad \int_0^{1/2} \frac{1}{\sqrt[3]{1+x^2}} dx \quad (\text{see Exercise 2})$$

11.10 Review: Concepts

Define or discuss each of the following.

- 1 Infinite sequence
- 2 Limit of an infinite sequence
- 3 Theorems on limits of sequences
- 4 Monotonic sequence
- 5 Sequence of partial sums of an infinite series
- 6 Convergent infinite series
- 7 Divergent infinite series
- 8 Geometric series
- 9 The harmonic series
- 10 The p -series
- 11 The Integral Test
- 12 Comparison tests

- 13 The Alternating Series Test
- 14 Absolute convergence
- 15 Conditional convergence
- 16 The Ratio Test
- 17 The Root Test
- 18 Power series
- 19 Radius of convergence
- 20 Interval of convergence
- 21 Differentiation and integration of power series
- 22 Taylor series
- 23 Maclaurin series
- 24 Binomial series
- 25 Approximations by series

Review: Exercises

In Exercises 1–6 find the limit of the sequence, if it exists.

$$1 \quad \left\{ \frac{\ln(n^2 + 1)}{n} \right\} \qquad 2 \quad \{100(0.99)^n\}$$

$$3 \quad \{10^n/n^{10}\} \qquad 4 \quad \left\{ \frac{1}{n} + (-2)^n \right\}$$

$$5 \quad \left\{ \frac{n}{\sqrt{n+4}} - \frac{n}{\sqrt{n+9}} \right\} \qquad 6 \quad \left\{ \left(1 + \frac{2}{n} \right)^{2n} \right\}$$

In Exercises 7–32, determine whether (a) a positive term series is convergent or divergent, or (b) a series which contains negative terms is absolutely convergent, conditionally convergent, or divergent.

$$7 \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n(n+1)(n+2)}}$$

$$8 \quad \sum_{n=0}^{\infty} \frac{(2n+3)^2}{(n+1)^3}$$

- 9 $\sum_{n=1}^{\infty} (-2/3)^{n-1}$
- 10 $\sum_{n=0}^{\infty} \frac{1}{2 + (1/2)^n}$
- 11 $\sum_{n=1}^{\infty} \frac{3^{2n+1}}{n5^{n-1}}$
- 12 $\sum_{n=1}^{\infty} \frac{1}{3^n + 2}$
- 13 $\sum_{n=1}^{\infty} \frac{n!}{\ln(n+1)}$
- 14 $\sum_{n=1}^{\infty} \frac{n^2 - 1}{n^2 + 1}$
- 15 $\sum_{n=1}^{\infty} (n^2 + 9)(-2)^{1-n}$
- 16 $\sum_{n=1}^{\infty} \frac{n + \cos n}{n^3 + 1}$
- 17 $\sum_{n=1}^{\infty} \frac{e^n}{n^e}$
- 18 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2 + 1}$
- 19 $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt[n]{n}}$
- 20 $\sum_{n=2}^{\infty} (-1)^n \frac{(0.9)^n}{\ln n}$
- 21 $\sum_{n=1}^{\infty} \frac{\sin(n5\pi/3)}{n^{5\pi/3}}$
- 22 $\sum_{n=2}^{\infty} (-1)^n \frac{\sqrt[3]{n-1}}{n^2 - 1}$
- 23 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n+1}$
- 24 $\sum_{n=1}^{\infty} (-1)^n \frac{2n+3}{n!}$
- 25 $\sum_{n=1}^{\infty} \frac{1 - \cos n}{n^2}$
- 26 $\frac{2}{1!} - \frac{2 \cdot 4}{2!} + \cdots + (-1)^{n-1} \frac{2 \cdot 4 \cdots (2n)}{n!} + \cdots$
- 27 $\sum_{n=1}^{\infty} \frac{(2n)^n}{n^{2n}}$
- 28 $\sum_{n=1}^{\infty} \frac{3^{n-1}}{n^2 + 9}$

- 29 $\sum_{n=1}^{\infty} \frac{e^{2n}}{(2n-1)!}$
- 30 $\sum_{n=1}^{\infty} \left(\frac{1}{3^n} - \frac{5}{\sqrt{n}} \right)$
- 31 $\sum_{n=2}^{\infty} (-1)^n \frac{\sqrt{\ln n}}{n}$
- 32 $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{\sqrt{1+n^2}}$

Use the Integral Test to determine the convergence or divergence of the series in Exercises 33–38.

- 33 $\sum_{n=1}^{\infty} \frac{1}{(3n+2)^3}$
- 34 $\sum_{n=2}^{\infty} \frac{n}{\sqrt{n^2-1}}$
- 35 $\sum_{n=1}^{\infty} n^{-2} e^{1/n}$
- 36 $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$
- 37 $\sum_{n=1}^{\infty} \frac{10}{\sqrt[3]{n+8}}$
- 38 $\sum_{n=5}^{\infty} \frac{1}{n^2 - 4n}$

Approximate the sums of the series in Exercises 39 and 40 to three decimal places.

- 39 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{(2n+1)!}$
- 40 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^2(n^2+1)}$

Find the interval of convergence of the series in Exercises 41–44.

- 41 $\sum_{n=0}^{\infty} \frac{n+1}{(-3)^n} x^n$
- 42 $\sum_{n=0}^{\infty} (-1)^n \frac{4^{2n}}{\sqrt{n+1}} x^n$
- 43 $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n} (x+10)^n$
- 44 $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2} (x-1)^n$

Find the radius of convergence of the series in Exercises 45 and 46.

- 45 $\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} x^n$
- 46 $\sum_{n=0}^{\infty} \frac{1}{(n+5)!} (x+5)^n$

In Exercises 47–52 find the Maclaurin series for $f(x)$ and state the radius of convergence.

- 47 $f(x) = (1 - \cos x)/x$ if $x \neq 0$ and $f(0) = 0$
- 48 $f(x) = xe^{-2x}$
- 49 $f(x) = \sin x \cos x$
- 50 $f(x) = \ln(2+x)$
- 51 $f(x) = (1+x)^{2/3}$
- 52 $f(x) = 1/\sqrt{1-x^2}$

53 Find a series representation for e^{-x} in powers of $x + 2$.

54 Find a series representation for $\cos x$ in powers of $x - \pi/2$.

55 Find a series representation for $\sqrt{x}$ in powers of $x - 4$.

Use infinite series to approximate the numbers in Exercises 56–60 to three decimal places.

56 $1/\sqrt[3]{3e}$

57 $\int_0^1 x^2 e^{-x^2} dx$

58 $\int_0^1 f(x) dx$, where $f(x) = (\sin x)/\sqrt{x}$ if $x \neq 0$ and $f(0) = 0$.

59 $\sqrt[5]{1.01}$

60 $e^{-0.25}$

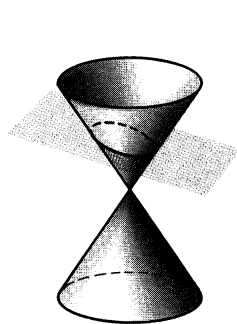
12 *Topics in Analytic Geometry*

Plane geometry includes the study of figures such as lines, circles, and triangles which lie in a plane. Theorems are proved by reasoning deductively from certain postulates. In *analytic* geometry, plane geometric figures are investigated by introducing a coordinate system and then using equations and for-

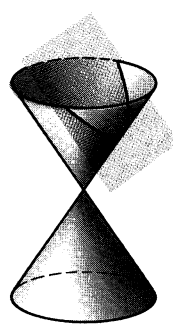
mulas of various types. If the study of analytic geometry were to be summarized by means of one statement, perhaps the following would be appropriate: "Given an equation, find its graph and, conversely, given a graph, find its equation." In this chapter we shall apply coordinate methods to several basic plane figures.

12.1 *Conic Sections*

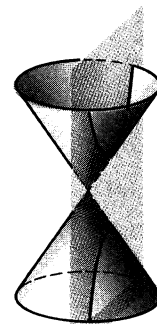
Each of the geometric figures to be discussed in this chapter can be obtained by intersecting a double-napped right circular cone with a plane. For this reason they are called **conic sections** or simply **conics**. If, as in (i) of Figure 12.1, the plane cuts entirely across one nappe of the cone and is not per-



(i) Ellipse



(ii) Parabola



(iii) Hyperbola

FIGURE 12.1

pendicular to the axis, then the curve of intersection is called an **ellipse**. If the plane is perpendicular to the axis of the cone, a **circle** results. If the plane does not cut across one entire nappe and does not intersect both nappes, as illustrated in (ii) of Figure 12.1, then the curve of intersection is a **parabola**. If the plane cuts through both nappes of the cone, as in (iii) of Figure 12.1, then the resulting figure is called a **hyperbola**.

By changing the position of the plane and the shape of the cone, conics can be made to vary considerably. For certain positions of the plane there result what are called **degenerate conics**. For example, if the plane intersects the cone only at the vertex, then the conic consists of one point. If the axis of the cone lies on the plane, then a pair of intersecting lines is obtained. Finally, if we begin with the parabolic case, as in (ii) of Figure 12.1, and move the plane parallel to its initial position until it coincides with one of the generators of the cone, we obtain a line.

The conic sections were studied extensively by the early Greek mathematicians, who discovered the properties that enable us to define conics in terms of points (foci) and lines (directrices) in the plane of the conic. Reconciliation of these definitions with the previous discussion requires proofs which we shall not go into here.

A remarkable fact about conic sections is that although they were studied thousands of years ago, they are far from obsolete. Indeed, they are important tools for present-day investigations in outer space and for the study of the behavior of atomic particles. It is shown in physics that if a particle moves under the influence of what is called an *inverse square force field*, then its path may be described by means of a conic section. Examples of inverse square fields are gravitational and electromagnetic fields. Planetary orbits are elliptical. If the ellipse is very “flat,” the curve resembles the path of a comet. The hyperbola is useful for describing the path of an alpha particle in the electric field of the nucleus of an atom. The interested person can find many other applications of conic sections.

12.2 Parabolas

Parabolas are very useful in applications of mathematics to the physical world. For example, it can be shown that if a projectile is fired and it is assumed that it is acted upon only by the force of gravity (that is, air resistance and other outside factors are ignored), then the path of the projectile is parabolic. Properties of parabolas are used in the design of mirrors for telescopes and searchlights, and in the construction of radar antenna. These are only a few of many physical applications.

Definition (12.1)

A **parabola** is the set of all points in a plane equidistant from a fixed point F (the **focus**) and a fixed line l (the **directrix**) in the plane.

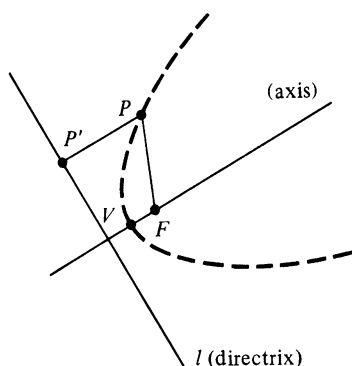


FIGURE 12.2

We shall assume that F is not on l , for otherwise the parabola degenerates into a line. If P is any point in the plane and P' is the point on l determined by a line through P which is perpendicular to l , then according to Definition (12.1), P is on the parabola if and only if $d(P, F) = d(P, P')$. A typical situation is illustrated in Figure 12.2, where the dashes indicate possible positions of P . The line through F , perpendicular to the directrix, is called the **axis** of the parabola. The point V on the axis, half-way from F to l , is called the **vertex** of the parabola.

To obtain a simple equation for a parabola, let us choose the y -axis along the axis of the parabola, with the origin at the vertex V , as illustrated in Figure 12.3. In this case, the focus F has coordinates $(0, p)$ for some real number $p \neq 0$, and the equation of the directrix is $y = -p$. By the Distance Formula, a point $P(x, y)$ is on the parabola if and only if

$$\sqrt{(x - 0)^2 + (y - p)^2} = \sqrt{(x - x)^2 + (y + p)^2}.$$

Squaring both sides gives us

$$(x - 0)^2 + (y - p)^2 = (y + p)^2,$$

or

$$x^2 + y^2 - 2py + p^2 = y^2 + 2py + p^2.$$

This simplifies to

$$x^2 = 4py.$$

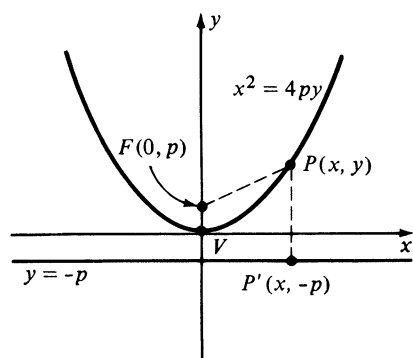


FIGURE 12.3

We have shown that the coordinates of every point (x, y) on the parabola satisfy $x^2 = 4py$. Conversely, if (x, y) is a solution of this equation, then by reversing the previous steps we see that the point (x, y) is on the parabola. If $p > 0$, the parabola **opens upward**, as in Figure 12.3, whereas if $p < 0$, the parabola **opens downward**. Note that the graph is symmetric with respect to the y -axis, since the solutions of the equation $x^2 = 4py$ are unchanged if $-x$ is substituted for x .

An analogous situation exists if the axis of the parabola is taken along the x -axis. If the vertex is $V(0, 0)$, the focus $F(p, 0)$, and the directrix has equation $x = -p$ (see Figure 12.4), then using the same type of argument we obtain the equation $y^2 = 4px$. If $p > 0$, the parabola opens to the right, whereas if $p < 0$, it opens to the left. In this case the graph is symmetric with respect to the x -axis.

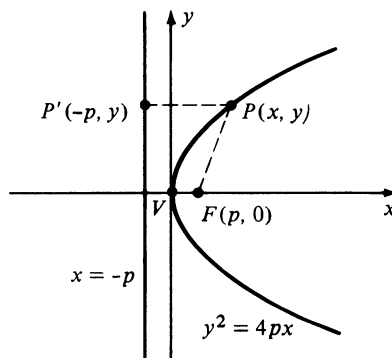


FIGURE 12.4

The following theorem summarizes our discussion.

Theorem (12.2)

The graph of each of the following equations is a parabola with vertex at the origin and having the indicated focus and directrix.

- (i) $x^2 = 4py$: focus $F(0, p)$, directrix $y = -p$.
- (ii) $y^2 = 4px$: focus $F(p, 0)$, directrix $x = -p$.

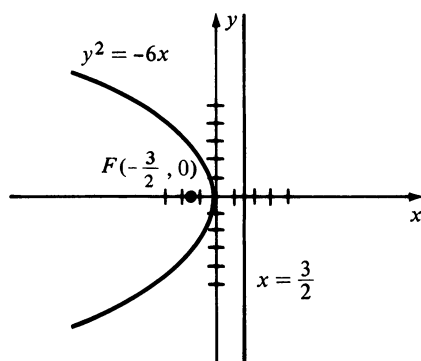


FIGURE 12.5

Example 1 Find the focus and directrix of the parabola having equation $y^2 = -6x$, and sketch the graph.

Solution The equation has form (ii) of Theorem (12.2) with $4p = -6$, and hence $p = -\frac{3}{2}$. Thus the focus is $F(p, 0)$, that is, $F(-\frac{3}{2}, 0)$. The equation of the directrix is $x = -p$, or $x = \frac{3}{2}$. The graph is sketched in Figure 12.5. ■

Example 2 Find an equation of the parabola that has its vertex at the origin, opens upward, and passes through the point $P(-3, 7)$.

Solution The general form of the equation is given by $x^2 = 4py$ (see (i) of Theorem (12.2)). If P is on the parabola, then $(-3, 7)$ is a solution of the equation. Hence we must have $(-3)^2 = 4p(7)$, or $p = \frac{9}{28}$. Substituting for p in (12.2) leads to the desired equation $x^2 = \frac{9}{7}y$, or $7x^2 = 9y$. ■

We may use a technique called **translation of axes** to extend the preceding discussion to the case in which the vertex of the parabola is not at the origin. Recall that if a and b are the coordinates of two points A and B , respectively, on a coordinate line l , then the distance between A and B is $d(A, B) = |b - a|$. If we wish to take into account the direction of l , then we use the **directed distance** $\overline{AB}$ from A to B where, by definition,

$$\overline{AB} = b - a.$$

Since $\overline{BA} = a - b$, we have $\overline{AB} = -\overline{BA}$. If the positive direction on l is to the right, then B is to the right of A if and only if $\overline{AB} > 0$, and is to the left of A if and only if $\overline{AB} < 0$. If C is any other point on l with coordinate c , it follows that

$$\overline{AC} = \overline{AB} + \overline{BC}$$

since $c - a = (b - a) + (c - b)$. We shall use this formula in the following discussion to develop formulas for translation of axes.

Suppose that $C(h, k)$ is an arbitrary point in an xy -coordinate plane. Let us introduce a new $x'y'$ -coordinate system with origin O' at C such that the x' - and y' -axes are parallel to, and have the same unit lengths and positive directions as, the x - and y -axes, respectively. A typical situation of this type